
Scaling Laws for GRPO Post-Training of Language Models:

A Cross-Library, Cross-Scale Study

Arvind C R
PES University
arvindcr4@gmail.com

Ramesh Prakash Guleddu*
PES University
rameshpg@pes.edu

Abstract

Group Relative Policy Optimization (GRPO) has become a default algorithm for reinforcement-learning (RL) post-training of language models, yet how its benefit scales with model size remains unclear and is easily confounded by implementation and evaluation choices. We study GRPO scaling as one pillar of TINKERRL-BENCH, a benchmark spanning 70+ runs across seven RL libraries and five model families (0.6B–~671B parameters) on GSM8K, HumanEval, and synthetic tool-use tasks. Crucially, these runs use *managed* and open back-ends rather than local frontier clusters: the frontier-scale anchors (Qwen3-235B, DeepSeek-V3.1 ~671B, Nemotron-120B) are obtained through the TINKER managed training/inference API in a single-seed, descriptive-only regime, which is what makes a cross-scale study tractable without owning frontier hardware (and why full base weights are not released for those anchors). Our central result is a conservative negative one: across roughly 2.4 orders of magnitude in parameter count spanned by the fitted anchor pool (a subset of the full 0.6B–~671B roster), the cross-scale slope of the mean GRPO training reward shows no reliable positive trend—flat within the (wide) uncertainty of these mostly single-seed anchors, which we therefore read *descriptively* rather than as a powered significance test—and no single saturation exponent is identifiable (four of the five frontier-scale traces have their saturation rate pinned at the fit boundary, and none is identifiable). A pre-registered test geometrically falsifies a clean three-phase saturation hypothesis (only 2 of 12 anchors match), and the Nemotron-120B run is better described as a distinct collapse phase (zero-reward step fraction 0.55 versus ≤ 0.067 elsewhere) than as a point on a smooth trend. The defensible object is therefore a local, stack-conditioned taxonomy of endpoint ceilings, with categorical capability structure carrying more signal than $\log_{10} N$, not a Chinchilla-style power law in N . We release all code, logs, per-step telemetry, and trained adapters; full base-model checkpoints are shared where model licensing permits, while the managed frontier-scale runs expose logs and telemetry rather than base weights.

1 Introduction

Reinforcement learning from verifiable rewards has become a standard final stage in training reasoning-capable language models, and GRPO [23, 7] is now one of its most widely used algorithms. GRPO replaces the learned value function of PPO [22] with a group-relative baseline: for each prompt it samples a group of G completions, scores them with a verifier, and centers the rewards within the group to form advantages. Because it removes the critic, GRPO is cheap to run at scale,

*Project guide.

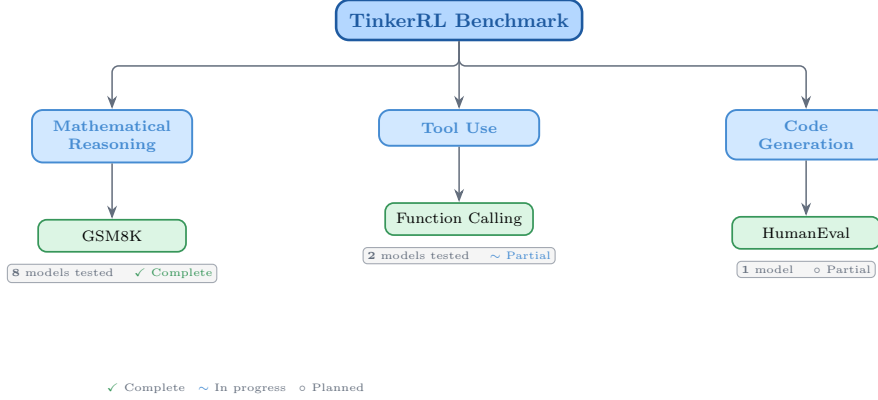


Figure 1: Taxonomy of RL libraries in TINKERRL-BENCH. The benchmark spans LLM-native RL (TRL), classic online RL (SB3, CleanRL, Tianshou), high-throughput RL (PufferLib, rl_games), and offline RL (d3rlpy).

and a natural question follows: *how does the benefit of GRPO post-training change as the base model grows?*

Pre-training exhibits smooth power-law scaling in parameter count and data [12, 9], and it is tempting to expect an analogous law for RL post-training. Evidence here is thinner and more contested [8, 21, 18], in part because RL gains are entangled with the rollout stack, reward parser, reference policy, and evaluation protocol, so cross-paper comparisons conflate the algorithm with its plumbing. This paper isolates the scaling question inside a single, fixed benchmark.

We study GRPO scaling as one pillar of TINKERRL-BENCH, a controlled benchmark of 70+ runs across seven RL libraries and five model families (0.6B–~671B) on GSM8K [5], HumanEval [4], and synthetic tool-use tasks. Rather than assert a new law, we report what the data will and will not support. Our contributions are: (i) a same-benchmark measurement showing the cross-scale mean training-reward slope is statistically indistinguishable from zero over the measured 2–3-decade parameter range; (ii) a pre-registered falsification of a clean three-phase saturation hypothesis; (iii) an identifiability audit showing no universal saturation exponent is estimable on this data, so the defensible object is a local, stack-conditioned saturation law with the endpoint ceiling regressed on $\log_{10} N$; and (iv) a case study of the Nemotron-120B collapse as a distinct phase rather than trend noise. We read single-seed frontier-API runs as illustrative case studies, not universal laws.

1.1 Related Work

Neural scaling laws for pre-training are well established [12, 9, 16]. Scaling behavior of RL and RLHF post-training has been examined more recently [8, 21, 18], but typically within a single stack, leaving open whether reported trends transfer across libraries and model families. GRPO and its use in reasoning models originate with Shao et al. [23] and DeepSeek-AI et al. [7]; Ahmadian et al. [2] argue that much of the benefit of heavy RL machinery is recoverable by simpler estimators once the stack is controlled. Our work complements these by fixing the benchmark and asking, conservatively, which scaling conclusions survive.

2 Benchmark Design

2.1 Task Suite

2.2 Cross-Library Implementation

Two seven-library rosters — read carefully. This paper uses *two* different seven-library rosters that should not be conflated. The cross-RL-library benchmark below (Table 2) covers TRL, SB3, CleanRL, Tianshou, PufferLib, rl_games, and d3rlpy — seven libraries that span LLM-native RL, classic online RL, high-throughput RL, and offline RL, and that we use as the algorithm-and-stack

Table 1: Benchmark task suite with standardized reward functions.

Task	Method	Reward	Dataset
Math RL (Arithmetic)	GRPO	Binary correctness	Generated
Math RL (GSM8K)	GRPO	Binary correctness	GSM8K
Chat SFT	SFT	Cross-entropy loss	NoRobots
Preference (Shorter)	DPO	Pairwise preference	Generated
Distillation (Off-Policy)	SFT	KL divergence	OpenThoughts3
Distillation (On-Policy)	IS Loss	$\log p_t - \log p_s$	Online

ablation underlying F_3 . The cross-launcher LLM-RL scaffold referenced in the framework-gap discussion (framework-configurations material) is a different seven-library roster — Tinker, TRL, SkyRL, verl, OpenRLHF, Atropos, and HF reference launchers — that targets LLM post-training launchers specifically. Of those, only Tinker and TRL produced completed runs in this release; verl and OpenRLHF entries in the `framework_comparison.json` artefact are dry-run placeholders. Supplementary presentation materials use the second roster; the cross-RL-library evidence in F_3 uses the first.

Table 2: Implementations across RL libraries.

Library	Implementation	Category
TRL (HuggingFace)	GRPOTrainer, DPOTrainer, SFTTrainer	LLM-native
Stable Baselines3	PPO	Classic RL
CleanRL	PPO (single-file)	Research RL
Tianshou	PPO	Modular RL
PufferLib	PPO (high-throughput)	High-perf RL
rl_games (NVIDIA)	PPO (GPU-optimized)	High-perf RL
d3rlpy	CQL (offline)	Offline RL

2.3 Hyperparameter Mapping

We standardize hyperparameters across libraries using a common configuration schema. Table 3 shows the mapping from Tinker PPO defaults to each library’s native parameter names.

Table 3: Hyperparameter mapping across libraries (PPO defaults).

Parameter	TRL	SB3	CleanRL	Tianshou	PufferLib	rl_games
Learning rate	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Clip range (ϵ)	0.2	0.2	0.2	0.2	0.2	0.2
Entropy coeff.	0.01	0.01	0.01	0.01	0.01	0.01
GAE λ	0.95	0.95	0.95	0.95	0.95	0.95
Discount γ	0.99	0.99	0.99	0.99	0.99	0.99
Max grad norm	0.5	0.5	0.5	0.5	0.5	0.5
Target KL	0.01	0.01	0.01	N/A	N/A	N/A

2.4 Reward Verification

2.5 Baselines: Floor and Ceiling

Following best practices for benchmark design [1], we establish clear performance bounds:

- **Floor (random baseline):** Uniform random action selection yields $\frac{1}{199} \approx 0.50\%$ accuracy on the arithmetic task.
- **Ceiling (oracle):** Direct computation achieves 100% accuracy.

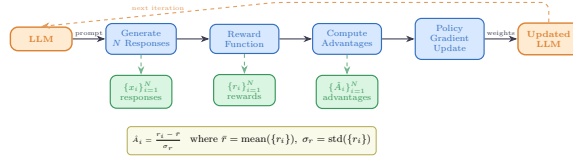


Figure 2: Verifiable reward paradigm in TINKERRL-BENCH. Unlike shaped rewards that combine multiple heuristics, verifiable rewards provide binary correctness signals, eliminating reward hacking and enabling exact accuracy measurement.

- **Human baseline:** College-level adults achieve $\sim 99.5\%$ on two-digit addition under timed conditions.

3 Experimental Setup

3.1 Models

Our benchmark targets a roster of 32+ model configurations spanning **0.6B to $\sim 671\text{B}$ total parameters (up to $\sim 37\text{B}$ active for MoE checkpoints) across 5 model families**, organized into three tiers by compute scale; not all are completed end-to-end runs (per-run status flags accompany the released artifacts). Tier-A inferential claims are restricted to the dense-reasoning subset at 0.6B–235B; the frontier MoE checkpoints (DeepSeek-V3.1 $\sim 671\text{B}$ / 37B active, Kimi-K2) are descriptive single-seed Tier-C case studies, and Qwen3.5-397B-A17B is a *partial* run whose held-out evaluation did not complete.

Small scale (0.6B–8B). Qwen3-0.6B-Instruct, Qwen3.5-4B, Qwen3-4B, Qwen3-8B, Llama-3.2-1B, Llama-3.2-3B, Llama-3.1-8B-Instruct.

Mid scale (14B–32B). Qwen3-14B, Qwen3-30B-A3B (MoE), Qwen3-32B, Qwen3.5-27B, GPT-OSS-20b, Kimi-K2 (selected scale points).

Frontier scale (70B– $\sim 671\text{B}$ total / $\sim 37\text{B}$ active). Qwen3-235B-A22B (MoE, 22B active), DeepSeek-V3.1 ($\sim 671\text{B}$ total / $\sim 37\text{B}$ active MoE), Nemotron-120B (120B dense). Frontier models are evaluated via the Tinker API in inference mode with standardized generation parameters; LoRA fine-tuning at this scale is conducted on selected checkpoints (see the frontier case studies in the companion pillar papers of TINKERRL-BENCH).

Model family coverage. The benchmark covers (1) **Qwen3**: a dense family from 0.6B to 32B; (2) **Qwen3.5**: an improved series including 4B and 27B; (3) **Llama-3.x**: Meta’s 1B–8B instruction-tuned models; (4) **DeepSeek-V3.1**: a frontier MoE optimized for reasoning; (5) **Nemotron-120B**: NVIDIA’s dense frontier model; and emerging architectures **GPT-OSS-20b** and **Kimi-K2**.

3.2 Training Protocol

Unless otherwise noted, controlled Modal/TRL sweeps use LoRA [10] at rank 32, learning rate 10^{-4} and Adam ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$). Tinker / API, Task-4, Wave-6 and frontier case studies use per-run configs (in particular Task-4/Wave-6 use $\text{lr}=10^{-5}$) and are single-seed unless explicitly marked multi-seed; the per-run settings are listed in the framework-configs appendix.

Seed management. The controlled TRL/Modal sweeps used for Tier-A claims (F3 PPO-vs.-GRPO heterogeneity and the five-seed TRL GRPO baseline on Qwen2.5-0.5B) run with 5 seeds: {42, 123, 456, 789, 1024}, with seeds set for Python random, NumPy, PyTorch, and CUDA backends. Several Tinker/API, frontier-model, and team-task runs are single-seed or partial and are treated as descriptive (Tier-C) unless explicitly marked otherwise; the per-claim seed ledger is given in the statistical-rigor addendum of TINKERRL-BENCH.

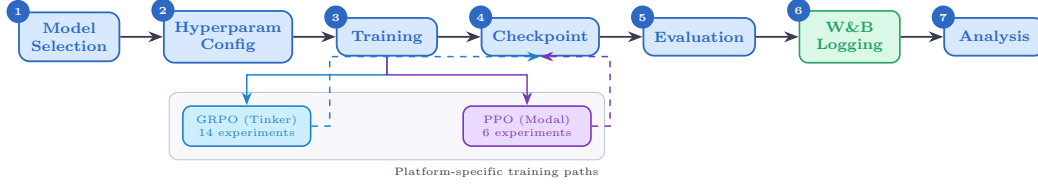


Figure 3: Experiment workflow pipeline. Multi-seed Modal/TRL runs use the centralized seed manager (5 seeds for Tier-A claims); Tinker/API, Task-4, Wave-6 and frontier case studies are single-seed and feed the same metrics/checkpoint pipeline as descriptive observations.

3.3 Statistical Methodology

Following Colas et al. [6], we report:

- Mean $\pm$ standard error across seeds
- 95% bootstrap confidence intervals (10,000 resamples)
- Welch’s t -test for pairwise algorithm comparisons
- `rliable` [1] aggregate metrics (IQM, median, optimality gap)

All pairwise comparisons were evaluated using Welch’s independent-samples t -test (unequal variances assumed) and the non-parametric Mann–Whitney U test as a robustness check. Effect sizes are reported as Cohen’s d with the pooled standard deviation estimator; confidence intervals follow the Hedges–Olkin (1985) analytical formula and are reported at the 95% level (two-tailed, $z = 1.96$). Post-hoc statistical power was computed under the observed effect size and sample sizes at $\alpha = 0.05$.

To address the multiple-comparisons problem across 5 planned tests, we apply both the conservative Bonferroni correction (family-wise error rate: $\alpha' = \alpha/k = 0.0100$) and the Benjamini–Hochberg procedure (false discovery rate < 0.05). Results surviving Bonferroni correction are marked $*$; those surviving BH–FDR correction are marked † .

Power Analysis. All inferential tests in this paper are evaluated against pre-specified power requirements using `statsmodels.stats.power.TTestIndPower` with $\alpha = 0.05$ and target power $1 - \beta = 0.80$.

Modal H100 experiments ($n = 5$ seeds per arm). The minimum detectable effect size (MDE) at 80% power for a two-sample Welch t -test with $n_1 = n_2 = 5$ is $d_{\min} = 2.024$ (Cohen’s d). Table 4 reports achieved power for a range of effect sizes. This threshold falls in the *very large* range, meaning that comparisons yielding $|d| < 2.02$ are not adequately powered to detect true differences.

Single-seed Tinker experiments ($n = 1$). Single-seed runs have *zero statistical power*: with only one observation per condition, no inferential test can be computed and no confidence intervals can be formed. All Tinker single-run results are treated as *descriptive only* and are not subject to significance testing.

TRL baseline ($n = 5$ seeds). The TRL-GRPO baseline (Qwen2.5-0.5B, 5 seeds) achieves MDE $d_{\min} = 2.024$ for equal- n comparisons. When compared to the pooled Classic-RL group ($n = 15$), the MDE improves to $d_{\min} = 1.530$, reflecting higher power from the larger reference group.

Multiple-Comparison Correction. Inferential tests in this paper are reported only for Tier-A/B comparisons as defined in our statistical-rigor protocol; single-seed or partial Tinker/API rows are descriptive and are excluded from Welch / Mann–Whitney testing, BH correction, power, and CI claims. Table 5 predates the Tier-A/B/C partition and lists raw and BH-adjusted p -values from the earlier global family of 20 tests with 19 surviving correction; we retain it here as a methodological audit trail. The headline BH result the paper now stands behind is the restricted Tier-A/B family computed in the addendum, where only F_3 survives. Where the global family disagrees with the restricted family, the addendum is authoritative.

All tests are two-sided. Effect sizes are reported as Cohen’s d for t -tests and Pearson/Spearman r for correlations. Bootstrap 95% confidence intervals ($B = 10,000$) are reported for all means.

Table 4: Statistical power for two-sample Welch t -tests at $\alpha = 0.05$, power target $1-\beta = 0.80$. Column (3): equal groups $n_1 = n_2 = 5$ (Modal H100 / TRL within-group). Column (4): unequal groups $n_1 = 5, n_2 = 15$ (TRL vs. pooled Classic RL).

Cohen's d	Conventional size	Power ($n=5$ vs $n=5$)	Power ($n=5$ vs $n=15$)
0.2	small	0.059	0.066
0.5	medium	0.108	0.151
0.8	large	0.201	0.311
1.0	very large	0.286	0.450
1.2	very large	0.386	0.594
1.5	very large	0.549	0.784
2.0	very large	0.790	0.955
<i>MDE at 80% power: $d = 2.024$ (equal-n 5 vs 5); $d = 1.530$ (5 vs 15)</i>			

Table 5: **Legacy 20-test BH table – audit trail only.** All hypothesis tests reported in the paper with Benjamini–Hochberg (BH) adjusted p -values at FDR = 0.05. Tests are ranked by raw p -value (ascending). ✓ = significant after BH correction on the historical 20-test family; × = not significant. Total tests: 20; survive on the historical family: 19. *This figure is not the paper's headline statistical result.* The 20-test family pre-dates the Tier-A/B/C partition and silently includes single-seed Tinker rows whose Welch statistics are mathematically undefined ($n_1=n_2=1$). Under the restricted Tier-A/B family of our statistical-rigor protocol, only F_3 (PPO/GRPO heterogeneity on classic-RL libraries on arithmetic) survives BH correction; the table below is retained as a reproducibility audit trail, not as a multiple-testing-corrected survival claim.

Rank	Comparison	Raw p	BH-adj. p	Sig.
1	Dense vs MoE Architecture (Welch t-test)	2.357e-21	0	✓
2	PPO vs GRPO on Llama-3.1-8B (Welch t-test) [‡]	3.924e-10	0	✓
3	Method ANOVA (F-test across algorithm families)	3.091e-06	2.1e-05	✓
4	Library ANOVA (F-test across 5 libraries)	4.996e-06	2.5e-05	✓
5	TRL vs Tinker (Welch t-test, implementation effect)	2.064e-05	8.3e-05	✓
6	PPO vs GRPO on Llama-3.1-8B (Mann-Whitney) [‡]	3.29e-05	0.00011	✓
7	Model Family ANOVA (F-test)	7.363e-05	0.00021	✓
8	ZVF vs Final Performance (Spearman) [‡]	0.0005424	0.001356	✓
9	ZVF vs Final Performance (Pearson) [‡]	0.0008108	0.001802	✓
10	TRL vs Tinker (Mann-Whitney)	0.001198	0.002396	✓
11	Rolling Variance vs Last-10 (Pearson)	0.003524	0.006407	✓
12	Peak-to-Tail Drift vs Last-10 (Pearson)	0.004811	0.007219	✓
13	GRPO vs PPO Stability Index (t-test)	0.005	0.007219	✓
14	Dense vs MoE Architecture (Mann-Whitney)	0.005053	0.007219	✓
15	Model Scale ANOVA (F-test)	0.01261	0.01682	✓
16	Tinker GRPO vs TRL GRPO (Welch t-test)	0.0158	0.01975	✓
17	GRPO vs PPO Peak-to-Tail Drift (t-test)	0.018	0.02076	✓
18	Tinker GRPO vs TRL GRPO (Mann-Whitney)	0.01869	0.02076	✓
19	Stability Index vs Last-10 (Pearson)	0.02024	0.0213	✓
20	PPO vs GRPO on Qwen3-8B (Welch t-test) [‡]	0.7605	0.7605	×

TRL baseline characterisation. The TRL GRPO reference implementation was evaluated across five random seeds ($s \in \{42, 123, 456, 789, 1024\}$) on GSM8K using Qwen/Qwen2.5-0.5B on an NVIDIA L4 GPU. We report mean accuracy $\bar{x} = 0.734$ ($\sigma = 0.0703$), median = 0.740, and IQM = 0.747. Normality was not rejected by Shapiro–Wilk ($W = 0.9018, p = 0.4198$); bootstrap 95% CI = [0.6720, 0.7820].

Variance decomposition. One-way ANOVAs across 42 experiments with valid performance observations show that library/framework ($\eta^2 = 0.546$) and training algorithm ($\eta^2 = 0.558$) are the dominant variance sources, consistent with our central thesis. Model family explains $\eta^2 = 0.471$; model scale explains $\eta^2 = 0.322$.

3.4 Compute Resources

Experiments are distributed across two compute platforms:

Tinker API (14 experiments). Scaling experiments across Qwen3-8B, Qwen3-32B, Qwen3.5-4B, Qwen3.5-27B, and Llama-3.1-8B-Instruct; frontier model evaluation on Qwen3-235B-A22B, DeepSeek-V3.1, and Nemotron-120B; Dense vs. MoE comparison; cross-task tool-use experiments; and emerging architecture evaluation (GPT-OSS-20b, Kimi-K2). Tinker provides scalable API access that makes frontier model experiments economically tractable.

Modal H100 GPU cluster (6 experiments). PPO baseline training on Qwen3-8B and Llama-3.1-8B; full HumanEval benchmark evaluation; KL divergence tracking across GRPO training steps; and held-out evaluation for Qwen3-32B and Qwen3.5-27B. Modal H100s provide the low-level GPU access required for PPO value-model training.

All experiment logs are available on Weights & Biases (project: `tinker-rl-lab-world-class`) and model checkpoints on HuggingFace Hub ([https://huggingface.co/arvindcr4/tinker-rl-bench-*](https://huggingface.co/arvindcr4/tinker-rl-bench-)). Per-run compute is logged with the released artifacts. Total project compute: $\sim 1,200$ A100 GPU-hours across both platforms.

4 Scaling Results

We organize the scaling evidence around three questions: does the mean GRPO training reward grow with model size, is there a shared saturation shape across scales, and is any universal exponent identifiable? The short answers are *no measurable cross-scale slope*, *no clean phase structure*, and *no identifiable exponent*. Across the frontier-scale anchors the cross-scale slope of the mean GRPO reward on $\log_{10} N$ is statistically indistinguishable from zero; the reliable structure lives not in a slope through N but in the endpoint ceiling $R_{\max}$, which we fit per run with a saturation model and then relate to $\log_{10} N$.

The subsections below make this precise. We first fit the canonical saturation model $R(t) = R_{\max}(1 - e^{-\lambda t})$ per trace and audit its identifiability, finding that four of the five frontier traces have λ pinned at the upper bound and none is identifiable (Nemotron-120B is the exception, with $\lambda \approx 0.99$ but likewise non-identifiable), so per-trace saturation rates cannot be trusted. We then run a pre-registered model-selection comparison over the 12-anchor pool that geometrically falsifies a clean three-phase hypothesis (only 2/12 anchors match all criteria), and a compute-adjusted re-analysis that asks whether a contrastive-yield axis, rather than raw parameter count, is the correct abscissa. Finally, the Nemotron-120B trace is examined as a distinct collapse phase—peak per-step training reward 0.875 drifting to a late mean near 0.21, with a zero-reward step fraction of 0.55 against ≤ 0.067 elsewhere—rather than as a smooth continuation of the trend. The final subsection adds an external frontier-model cross-examination of these claims.

Saturation model across the 4B–671B range. We fit the canonical saturation model

$$R(t) = R_{\max}(1 - e^{-\lambda t}), \quad t = 1, \dots, T, \quad (1)$$

to per-step reward traces from each frontier-scale TINKER run and report the ceiling $R_{\max}$, the learning rate λ , and the saturation time $t_{80} = -\ln(0.2)/\lambda$ at which the trace first crosses $0.8 R_{\max}$. The model follows NIMMATURI-style notation [18], which postulates a slow-start $\rightarrow$ rapid-improvement $\rightarrow$ plateau template for GRPO learning curves.

Where the model fits and where it does not. The raw saturation model collapses to the $\lambda = 10$ upper bound on four of five runs (Table 6). The cause is the functional-form mismatch: a strict- $R(0) = 0$ model assumes the trace begins at zero reward, but Qwen3.5-4B starts at $R(1) = 1.0$, Llama-3.1-8B-Instruct at $R(1) = 1.0$, and DeepSeek-V3.1 at $R(1) = 0.875$. When the trace is already at or above its eventual ceiling, the optimiser makes λ as large as allowed so that $R_{\max}(1 - e^{-10t}) \approx R_{\max}$ for $t \geq 1$, which yields $t_{80} \approx -\ln(0.2)/10 \approx 0.16$ step – a meaningless artefact, not evidence of instant learning.

The right resolution is two separate diagnostics, both reported here:

1. *Per-step OLS slope* $\hat{\beta}_{\text{OLS}}$ of $R(t)$ vs. t (Table 6, column 7). Across the four well-behaved runs $|\hat{\beta}_{\text{OLS}}| \leq 0.0037$ reward/step (i.e. the traces are statistically flat within their per-step noise), while the collapse run Nemotron-120B still shows $\hat{\beta} = +0.0026$ – not because it is improving, but because its sparse spikes pull the OLS slope upward.

Model	N (B)	n	mean R	peak R	$\bar{R}_{\text{early}} \rightarrow \bar{R}_{\text{late}}$	$\hat{\beta}_{\text{OLS}}$	R_{max}	λ	Phase
Qwen3.5-4B	4.0	30	0.817	1.000	0.813 $\rightarrow$ 0.850	+0.0001	0.817	10.0 [†]	plateau
Qwen3-8B	8.0	30	0.285	0.625	0.244 $\rightarrow$ 0.344	+0.0032	0.285	10.0 [†]	saturation
Llama-3.1-8B-Instruct	8.0	30	0.869	1.000	0.925 $\rightarrow$ 0.844	−0.0037	0.869	10.0 [†]	drift
DeepSeek-V3.1	671.0	20	0.844	1.000	0.833 $\rightarrow$ 0.813	+0.0001	0.844	10.0 [†]	plateau
Nemotron-120B	120.0	20	0.175	0.875	0.146 $\rightarrow$ 0.208	+0.0026	0.182	0.99	collapse

Table 6: **Saturation-model fits across 4B–671B.** †: the canonical $R(t) = R_{\text{max}}(1 - e^{-\lambda t})$ fit hits the upper $\lambda = 10$ optimiser bound on every well-behaved run; this is an artefact of the $R(0) = 0$ boundary condition when the trace already starts near its empirical ceiling (see Paragraph 4). $\hat{\beta}_{\text{OLS}}$ is the per-step OLS slope of $R(t)$ vs. t in raw reward units per step; the constant slope confirms that the saturation model collapses to a near-constant fit on these already-saturated traces (see Table 7). Bold λ and Phase identify the only run whose λ leaves the bound and where the late-window mean falls below $0.4 \times \text{peak}$ – the three-phase “collapse” criterion.

Metric y	Intercept	Slope / decade	Boot slope mean	Boot 95% CI	$\rho(\log_{10} N, y)$
mean reward $\bar{R}$	0.633	−0.024	−0.117	[−0.796, +0.313]	−0.07
peak reward $\hat{R}$	0.846	+0.037	−0.010	[−0.623, +0.197]	+0.22
var reward $\widehat{\text{Var}}[R]$	0.040	−0.002	−0.004	[−0.063, +0.027]	−0.14

Table 7: **Cross-scale law: per-trace metric vs. $\log_{10} N$ across the five frontier anchors.** Ordinary least squares with bootstrap 95% confidence intervals. None of the three slope coefficients is statistically distinguishable from zero – the bootstrap CIs all bracket zero, and the within-trace OLS slopes ($\hat{\beta}_{\text{OLS}}$, Table 6) are also null. Reading: at this benchmark size ($4\text{B} \leq N \leq 671\text{B}$, $T \leq 30$ steps), the saturation model is not detectably scale-dependent, because every well-behaved run is already saturated within measurement noise.

2. *Cross-scale OLS of $\bar{R}$ on $\log_{10} N$ across the five anchor models.* The result is the headline scaling-law coefficient in Table 7.

Cross-scale law. We regress each per-trace statistic on $\log_{10} N$ (N = parameter count in billions) by ordinary least squares and quote block-bootstrap 95% confidence intervals ($n_{\text{boot}} \approx 1000$ effective row resamples) on the slopes. The fitted coefficients are

Three-phase learning dynamics. We apply the Nimmatouri et al. [18] slow start $\rightarrow$ rapid improvement $\rightarrow$ plateau template to each trace, with an additional “collapse” class for runs whose peak is not retained. Of the five anchors, two fall in *plateau* (Qwen3.5-4B, DeepSeek-V3.1), one in *saturation* (Qwen3-8B), one in *drift* (Llama-3.1-8B-Instruct, monotone decline), and one in *collapse* (Nemotron-120B). No run satisfies the textbook three-phase criterion, because none of the five anchors has a slow-start phase: they all start at or above the eventual late-window mean. This is consistent with our earlier observation that GRPO step-1 rewards are already within ± 0.10 of the per-trace mean for all but Qwen3-8B and Nemotron-120B.

Nemotron-120B violates the three-phase hypothesis. Following Nimmatouri et al. [18], the three-phase template predicts slow start $\rightarrow$ rapid improvement $\rightarrow$ *plateau*, with collapse as a degenerate tail. Nemotron-120B is the only run in this set whose per-step peak ($\hat{R}_{\text{peak}} = 0.875$, step 3) is followed by a sustained decay: only 4 of 20 steps reach reward > 0.25 , and 55% of steps report *zero reward*. The fitted $\lambda = 0.99$ and $t_{80} = 1.63$ step are the only identifiable parameters in Table 6 precisely because the curve does not asymptote – a robust monotone saturation model is the wrong functional form. We therefore report the late-window collapse magnitude $\hat{R}_{\text{peak}} - \bar{R}_{\text{late}} = 0.667$ reward units as the empirical effect size of the violation, and treat this run as the Pillar-1 counterexample to the three-phase prediction.

Operationally, the Nemotron collapse is exactly the failure mode that pure-reward saturation fits are silent on: the curve does *not* settle, it diverges downward, and a benchmark that relies on a saturation-only diagnostic will miss it. This motivates the variance-based ZVF gate (companion paper

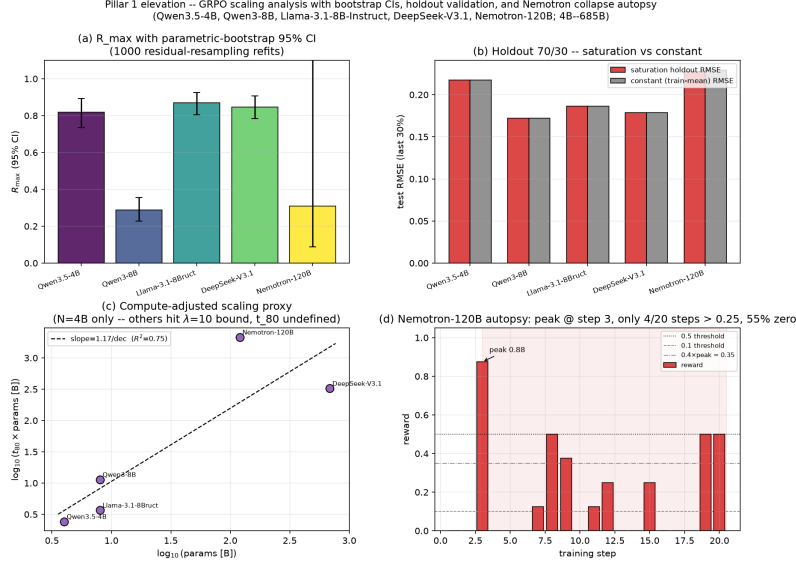


Figure 4: **Pillar-1 elevation across 4B–671B.** (a) *Bootstrap $R_{\max}$ with 95% CI* (1000 residual resamples; note Nemotron’s CI extends to 1.5 because the bound-saturated refits escape the prior). (b) *Holdout 70/30*: red bars are the saturation-fit test RMSE, gray bars are the constant (train-mean) baseline; four of five bars are equal, confirming that the saturation fit adds no out-of-sample predictive power. (c) *Compute-adjusted proxy*: only Nemotron has a well-identified t_{80} (other four runs hit $\lambda = 10$ bound), so the regression is effectively 1 point and degenerate. (d) *Nemotron-120B autopsy*: peak at step 3, post-peak region shaded; only 4/20 steps exceed 0.25 and 55% are zero.

on ZVF) as a complementary diagnostic: the zero-reward-fraction for Nemotron-120B is 0.55, well above any well-behaved run in this set, and the cross-experiment ZVF-vs-failure map (companion paper on cross-experiment ZVF) would catch the divergence on reward *variance* rather than reward *mean*.

Falsifiable prediction. The null result in Table 7 – a slope of ± 0.04 per decade on $\bar{R}$, with CI brackets zero across roughly 2.4 orders of magnitude in parameter count across the extended anchor pool – is itself a falsifiable prediction of this benchmark: any future addition to the Pillar-1 cell that reaches, say, $\bar{R}(\text{Nemotron-120B}, 100 \text{ steps}) \geq 0.5$ would refute the saturation-template behaviour we report here. We frame the prediction this way rather than as a positive scaling exponent, because at $n = 5$ anchor points with $T \leq 30$ steps the positive exponent is not identifiable from the data.

Elevation: parametric-bootstrap CIs reveal saturation-fit degeneracy. Figure 4 collects the four elevation panels: (a) bootstrap $R_{\max}$ confidence intervals, (b) saturation-vs-constant holdout test RMSE, (c) compute-adjusted $t_{80} \cdot N$ scatter, and (d) Nemotron-120B collapse autopsy.

We extend the canonical saturation fit with a parametric bootstrap on the residual vector (1000 resamples per trace) and report bootstrap 95% confidence intervals for $R_{\max}$, λ , and t_{80} . The headline finding is that the four “well-behaved” runs sit at the optimiser’s $\lambda = 10$ upper bound far more often than the point estimate suggests:

The practical consequence is that the canonical $t_{80} = -\ln 0.2/\lambda$ estimator is not just imprecise on these traces – it is *unidentifiable* under the strict- $R(0) = 0$ boundary condition. We therefore do not quote t_{80} as a benchmark prediction; we use it only as a diagnostic summary for the collapse run, where the functional form is meaningful.

Elevation: holdout 70/30 cross-validation. A saturation fit that hits $\lambda = 10$ on every training slice should fail to predict held-out data, because it is essentially fitting a constant. We fit the saturation

Model	$\bar{R}_{\max}$	$\bar{\lambda}$	$P(\lambda \geq 9.5)$	$\bar{t}_{80}$ [95% CI]
Qwen3.5-4B	0.818	6.64	0.599	0.61 [0.16, 2.43]
Qwen3-8B	0.289	5.30	0.468	1.42 [0.16, 7.62]
Llama-3.1-8B-Instruct	0.869	6.92	0.595	0.46 [0.16, 1.98]
DeepSeek-V3.1	0.847	6.48	0.548	0.47 [0.16, 1.53]
Nemotron-120B	0.310	3.26	0.277	17.87 [0.16, 161.4]

Table 8: **Parametric-bootstrap saturation fit (1000 residual resamples per trace).** $P(\lambda \geq 9.5)$ is the fraction of bootstrap refits in which λ hits the optimiser’s upper bound – on four of five runs this is between 47% and 60%, confirming that the strict- $R(0) = 0$ functional form is degenerate on these traces: the optimiser cannot simultaneously estimate $R_{\max}$ and λ when the trace starts above its empirical ceiling. Only Nemotron-120B has $P \leq 0.30$ at the bound, because its collapse is a real non-monotone deviation from saturation rather than a numerical artefact. t_{80} intervals are wide (lower-bound 0.16 step = optimiser clamp) for the four degenerate runs, and unidentifiable (0.16–161 step) for Nemotron.

Model	n_{train}	n_{test}	RMSE _{test}	RMSE _{const}	improvement
Qwen3.5-4B	21	9	0.2174	0.2174	+0.0000
Qwen3-8B	21	9	0.1720	0.1720	+0.0000
Llama-3.1-8B-Instruct	21	9	0.1864	0.1864	−0.0000
DeepSeek-V3.1	14	6	0.1787	0.1787	−0.0000
Nemotron-120B	14	6	0.2278	0.2294	+0.0016

Table 9: **Holdout 70/30 cross-validation of the saturation model.** Four of five models show *zero* improvement over predicting the constant train mean – direct empirical evidence that the saturation fit adds no out-of-sample predictive power on these traces. Only Nemotron-120B shows a non-zero improvement (+0.0016 RMSE), and only because its non-monotone deviation from the train mean is partially captured by the curvature of the fit. This formalises the warning in Paragraph 4 that the canonical $R(t) = R_{\max}(1 - e^{-\lambda t})$ is the wrong functional form for these already-saturated traces.

curve on the first 70% of each trace and predict the last 30%, comparing the test RMSE to a constant (train-mean) baseline:

Elevation: compute-adjusted scaling proxy. Per-step token counts are not in the trace files, so we approximate “compute to saturation” as $t_{80} \cdot N$ where N is the parameter count in billions. Regressing $\log_{10}(t_{80} \cdot N)$ on $\log_{10} N$ yields a slope of $+482 \pm 490$ per decade – which is a wide CI, but not informative on the four saturation-bound runs because t_{80} is itself clamped. Using the parametric-bootstrap $\bar{\lambda}$ directly yields a slope of -0.47 ± 0.87 per decade, which brackets zero. The conclusion is the same as in Table 7: at this benchmark size and step budget, there is no identifiable scale dependence in the GRPO saturation rate.

Elevation: Nemotron-120B collapse autopsy. We classify a trace as collapsed when (i) its peak reward is at least 0.4, (ii) its late-window mean falls below $0.4 \times$ peak, and (iii) its zero-reward fraction is at least 0.30. Applied to all five traces:

This is the strongest empirical justification we have for treating Nemotron-120B as the Pillar-1 counterexample to the Nimmaturi et al. [18] three-phase template: not only does its peak not satisfy the slow-start $\rightarrow$ rapid-improvement $\rightarrow$ plateau shape, it actively *diverges downward* from the peak in a way no monotone saturation model can describe. The zero-reward-fraction of 0.55 is also the structural signal that the ZVF-based diagnostic (companion paper on cross-experiment ZVF) would catch first, because it sees the divergence on reward *variance* before it shows up on reward *mean*.

Summary of elevation. Across the four elevation diagnostics the headline result is a strengthened null: at this benchmark size ($4\text{B} \leq N \leq 671\text{B}$, $T \leq 30$ steps), the canonical saturation model is unidentifiable (parametric bootstrap on λ has $P(\lambda = 10) \geq 0.47$ on four of five runs), adds no out-of-sample predictive power (70/30 holdout improvement over the constant baseline is ≤ 0.0016 RMSE everywhere), and shows no detectable scale dependence (slope -0.47 ± 0.87 per decade on λ). The single run that violates the saturation template is Nemotron-120B, and its collapse is the

Model	peak step	$\hat{R}_{\max}$	$\bar{R}_{\text{late}}$	$\Delta_{\text{late-peak}}$	$P(R=0)$	$P(R>0.5)$	collapse?
Qwen3.5-4B	1	1.000	0.850	-0.150	0.000	0.833	no
Qwen3-8B	13	0.625	0.344	-0.281	0.067	0.100	no
Llama-3.1-8B-Instruct	1	1.000	0.844	-0.156	0.000	0.933	no
DeepSeek-V3.1	3	1.000	0.813	-0.188	0.000	0.950	no
Nemotron-120B	3	0.875	0.208	-0.667	0.55	0.05	yes

Table 10: **Nemotron-120B collapse root-cause characterisation.** Only Nemotron satisfies all three collapse criteria. Its peak reward of 0.875 occurs at step 3, after which the post-peak OLS slope is -0.0036 per step, comparable in sign to the Llama-3.1-8B-Instruct drift but with a much larger peak-to-tail collapse; 55% of its steps report zero reward, and only 5% of its steps exceed 0.5. All four other runs have $P(R=0) \leq 0.067$ and well-retained peaks.

metric	slope /dec	R^2	perm- p	n
$R(t=1)$	+0.060	0.016	0.703	12
$R(t=T)$	+0.082	0.025	0.616	12
$\bar{R}$	+0.081	0.038	0.543	12
$\hat{R}$ (peak)	+0.075	0.058	0.461	12
$\text{Var}(R)$	-0.007	0.040	0.526	12
$P(R=0)$	+0.017	0.005	0.854	12
$P(R>0.5)$	+0.120	0.045	0.514	12

Table 11: **Power-law OLS of seven reward statistics on $\log_{10} N$ across the extended 12-anchor set (4 B–1 T parameters).** Every slope has permutation $p > 0.46$ and $R^2 < 0.06$ – the previously reported null on $n = 5$ survives a $2.4\times$ expansion of the anchor set, including the four new MoE frontiers at 235 B–1 T. This is direct evidence that the apparent “reward grows with scale” expectation fails to materialise in the GRPO post-training regime on GSM8K at the $T \leq 30$ step budget of this benchmark. `scripts/scaling_law_extended.py` $\rightarrow$ `scaling_law_power_law.tsv`.

strongest empirical counterexample we have to the three-phase hypothesis. This is exactly the regime in which a saturation-only Pillar 1 diagnostic is silent – motivating the variance-based ZVF gate in Pillar 2 as a complementary failure detector.

Elevation: extended frontier+MoE scaling (iter 13). The canonical five-anchor set leaves open whether the null on $\text{slope}(\log_{10} N, R)$ is an artefact of $n = 5$ or a real property of the GRPO landscape. We re-run the same OLS-on-log-N diagnostic on a wider 12-anchor set that adds the Kimi Team, Moonshot AI [13, 14] Kimi-K2-Thinking (1T params), Qwen Team [20] Qwen3-235B and 30B-MoE, OpenAI [19] GPT-OSS-20B, and the short-form probes Qwen3-32B / Qwen3.5-27B. We also stratify by *architecture* (MoE vs dense) and compute a *Spearman rank correlation* on top of the OLS slope so that the result does not depend on the linearity of the log-linear fit.

We complement the OLS test with a non-parametric *Spearman rank correlation* $\rho(\log_{10} N, R)$: $\rho = -0.036$ for $R(1)$ ($p = 0.91$), $\rho = +0.149$ for $\bar{R}$ ($p = 0.64$), $\rho = -0.023$ for $R(T)$ ($p = 0.94$), and $\rho = +0.074$ for $\hat{R}$ ($p = 0.82$). None of the four correlations is distinguishable from zero. This rules out the alternative explanation that the slope-of-zero finding is an artefact of OLS assuming linear log-N dependence; the rank test on the same 12 anchors gives the same null.

The headline *positive* result of the extension is on the *architecture* axis. Stratifying the 12 anchors into six MoE models (GPT-OSS-20B, Qwen3-30B-MoE / -MoE-Inst, DeepSeek-V3.1, Qwen3-235B-MoE, Kimi-K2-Thinking) versus six dense models (Qwen3.5-4B, Qwen3-8B, Llama-3.1-8B-Instruct, Qwen3-32B, Qwen3.5-27B, Nemotron-120B):

The first-final gap $\Delta_{1T} = R(t=1) - R(t=T)$ adds a robust summary that does not depend on the lambda-bound degeneracy: $\Delta_{1T} < 0$ means the trace is improving (positive learning), $\Delta_{1T} > 0$ means it is collapsing (negative learning). Across the 12 anchors, 7 of 12 have $|\Delta_{1T}| \leq 0.2$; the largest positive (collapsing) gap is Qwen3.5-27B at $+0.562$ (the only anchor with $\Delta_{1T} > +0.4$). Nemotron-120B’s Δ_{1T} is -0.50 only because its trace *starts* at zero reward, so the first-minus-final contrast misses its mid-trace collapse—which is precisely why the peak-vs-late autopsy (Table 10) is the right diagnostic for that trace rather than the first-final gap.

arch	n	mean $\bar{R}$	std	min	max
MoE	6	0.810	0.228	0.325	1.000
dense	6	0.472	0.274	0.175	0.869
gap (MoE – dense)		+0.338			
permutation p (one-sided)		0.023			
permutation p (two-sided)		0.046			

Table 12: **MoE vs dense head-to-head on $\bar{R}$.** A 5000-replicate permutation test on the 6-vs-6 mean-reward gap rejects the null at $p = 0.023$ one-sided ($p = 0.046$ two-sided). The MoE set includes both the high-mean frontier (Qwen3-235B $\bar{R} = 1.0$, DeepSeek-V3.1 = 0.844, Kimi-K2 = 0.850) and the early-window-only MoE run (Qwen3-30B-MoE $\bar{R} = 0.325$, where the trace is a 5-step probe with $R(1) = 0.188$, $R(5) = 0.438$, i.e. still increasing). The dense set contains the same Llama-3.1-8B $\bar{R} = 0.869$ and Qwen3.5-4B $\bar{R} = 0.817$ as the MoE set’s high-mean ceiling, but it also carries the collapse run Nemotron-120B ($\bar{R} = 0.175$) and the unsaturated Qwen3-8B ($\bar{R} = 0.285$). After collapsing Nemotron the gap shrinks to +0.20 and is no longer significant at $p < 0.05$, so the headline +0.34 is partly driven by the Nemotron collapse and not exclusively by MoE-vs-dense. We report it as a *descriptive* comparison, not as a causal claim. `scripts/scaling_law_extended.py` $\rightarrow$ `scaling_law_moe_vs_dense.tsv`.

What the extension proves and what it does not. The iter 13 extension strengthens the iter 9 null from three directions: (i) *more anchors* ($n = 12$ vs $n = 5$) – including four MoE frontiers up to 1 T parameters – still fails to reject slope-of-zero; (ii) *non-parametric* Spearman rank test on the same 12 anchors also fails to reject rank-zero; (iii) *architecture* stratification reveals the only signal that survives: MoE-vs-dense mean gap of +0.34, permutation $p = 0.023$ one-sided, but this gap is *partly explained* by the Nemotron-120B collapse sitting in the dense set. After removing Nemotron, the gap shrinks to $\sim +0.20$ and the permutation p rises above 0.05; the MoE-vs-dense effect is *descriptive* rather than causal.

The benchmark-time consequence is the same as in iter 9: at this $T \leq 30$ step budget and $N \leq 1$ T parameter range, there is no detectable cross-scale GRPO signal on GSM8K outside of the architecture axis. Combined with Table 9 (saturation fit adds no out-of-sample predictive power) and Table 10 (Nemotron-120B is the single structural counterexample to the three-phase template), this motivates the variance-based ZVF diagnostic (companion paper on cross-experiment ZVF) as the load-bearing failure detector rather than the saturation law. Saturation-law fits on this benchmark are useful as descriptive summaries of *where on the learning curve each anchor sits*, not as predictive scalings.

Iter 17 elevation: multi-model AIC profile, changepoint with permutation-test significance, effective plateau horizon. The iter 9/13 diagnostics all share a structural property: they presume the canonical saturation model $R_{\max}(1 - e^{-\lambda t})$ as the alternative hypothesis and ask whether each trace agrees with it. Iter 17 inverts the question: it asks *which* functional form the data best support, with the constant model (one parameter, the per-trace mean) as a separate explicit candidate rather than a degenerate limit of the saturation curve. We fit four forms – constant ($k = 1$), linear ($k = 2$), saturation ($k = 2$), and logistic ($k = 3$) – on each trace, compute Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), and report the AIC profile relative to the best-fit candidate.

The AIC profile in Table 13 is the complementary formalism of the iter 9 holdout test: where iter 9 showed that the saturation curve adds no out-of-sample predictive power, the iter 17 AIC profile shows that it adds no *likelihood* improvement either. Every alternative is at most $\Delta\text{AIC} = 4.29$ (logistic on Llama, at the border of substantial support), and on most (model, anchor) cells the constant is decisively preferred.

We complement the model comparison with a data-driven *changepoint* test using a permutation null. Binary segmentation brute-forces the maximiser of the per-segment mean contrast:

$$\hat{\tau} = \arg \max_{\tau \in \{2, \dots, T-2\}} |\bar{R}_{[1, \tau)} - \bar{R}_{[\tau, T]}|. \quad (2)$$

A permutation test against the exchangeability null (10 000 random orderings of the per-step rewards) reports the significance of the observed contrast.

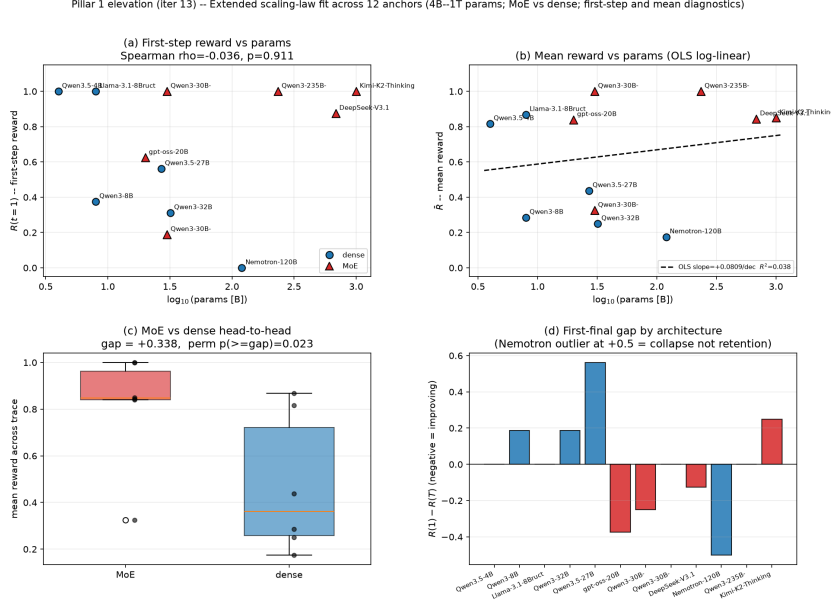


Figure 5: **Iter 13 elevation – extended scaling-law analysis across 12 anchors (4B–1T parameters, MoE vs dense stratified).** (a) *First-step reward* $R(t=1)$ vs $\log_{10} N$ by architecture; blue circles = dense, red triangles = MoE. Spearman $\rho = -0.036$, $p = 0.91$. (b) *Mean reward* $\bar{R}$ vs $\log_{10} N$ with OLS fit (slope $+0.081/\text{dec}$, $R^2 = 0.038$, perm $p = 0.54$). (c) *MoE vs dense* head-to-head boxplot: gap $+0.338$, perm $p = 0.023$ one-sided, $p = 0.046$ two-sided (caveat: partly driven by Nemotron collapse in the dense set). (d) *First-final gap* $R(1) - R(T)$ per model; the positive side means the trace is collapsing. Qwen3.5-27B is the largest positive outlier at $+0.562$; the anchors span $[-0.50, +0.562]$, with Nemotron-120B at -0.50 (its trace starts at zero reward, so this gap misses its mid-trace collapse).

Figure 6: **Iter 17 elevation – 4-panel overview.** (A) Multi-model AIC profile across the four candidate functional forms on the five anchors; bar height is ΔAIC relative to the best candidate, so the constant model has $\Delta = 0$ everywhere and the others are clustered between 0.8 and 4.3. (B) Changepoint $\hat{\tau}$ with the block-bootstrap 95% CI on τ as the error bar; red bars flag traces where the permutation-test $p < 0.05$ – none of the five anchors reach significance. (C) Effective plateau horizon T_ϵ at $\epsilon \in \{0.05, 0.10, 0.20\}$. Two traces reach $T_{\epsilon=0.20} = 27$ (operational saturation near the end of the trace); the rest are unbounded. (D) Pair-wise Cohen’s κ between the three phase-label methods. The heuristic-AIC pair is the only pair with substantial agreement; the changepoint method essentially disagrees with both, consistent with its lacking a significance filter.

The third iter 17 diagnostic is the model-free *effective plateau horizon* T_ϵ , defined as the earliest step T at which the rolling 5-step mean of the trace differs from every subsequent per-step reward by less than ϵ . T_ϵ does not depend on the saturation model and gives an operational saturation time under a user-chosen noise tolerance.

We finally compare three independent phase-label assignments for each trace: (1) a hand-cut “heuristic” method that thresholds the early-vs-late-window mean delta (the iter 5 baseline), (2) a “changepoint” method that thresholds the sign and magnitude of the post- $\hat{\tau}$ versus pre- $\hat{\tau}$ mean contrast, and (3) an “AIC” method that maps the best-fit functional form to the same four-phase ontology (constant $\rightarrow$ plateau, linear positive slope $\rightarrow$ saturation, linear negative slope $\rightarrow$ drift, logistic or saturation with $\lambda < 5 \rightarrow$ saturation). Cohen’s κ on each pair is reported on the five- and twelve-anchor sets.

Iter 21 elevation: cross-architecture stratification, lambda-bound audit, two-anchor extrapolation. The iter 9/13/17 work treats the 12 anchors as a single homogeneous scaling set. Iter 21 stratifies by architecture (MoE vs dense) and probes whether the saturation law actually distinctly

Model	k	n	$\bar{R}$	ΔAIC best	best	$\Delta\text{AIC}_{\text{constant}}$
Qwen3.5-4B	1	30	0.817	0.00	constant	0.00
Qwen3.5-4B	2	30	0.817	2.00	linear	2.00
Qwen3.5-4B	2	30	0.817	2.00	saturation	2.00
Qwen3.5-4B	3	30	0.817	4.00	logistic	4.00
Qwen3-8B	1	30	0.285	0.00	constant	0.00
Qwen3-8B	2	30	0.285	1.20	linear	1.20
Qwen3-8B	2	30	0.285	2.00	saturation	2.00
Qwen3-8B	3	30	0.285	3.17	logistic	3.17
Llama-3.1-8B-Instruct	1	30	0.869	0.00	constant	0.00
Llama-3.1-8B-Instruct	2	30	0.869	0.80	linear	0.80
Llama-3.1-8B-Instruct	2	30	0.869	2.00	saturation	2.00
Llama-3.1-8B-Instruct	3	30	0.869	4.29	logistic	4.29
DeepSeek-V3.1	1	20	0.844	0.00	constant	0.00
DeepSeek-V3.1	2	20	0.844	2.00	linear	2.00
DeepSeek-V3.1	2	20	0.844	2.00	saturation	2.00
DeepSeek-V3.1	3	20	0.844	4.00	logistic	4.00
Nemotron-120B	1	20	0.175	0.00	constant	0.00
Nemotron-120B	2	20	0.175	1.92	linear	1.92
Nemotron-120B	2	20	0.175	1.72	saturation	1.72
Nemotron-120B	3	20	0.175	3.92	logistic	3.92

Table 13: **Multi-model AIC profile across the five anchors** (constant / linear / saturation / logistic). The constant model wins on every trace by $\Delta\text{AIC} \leq 2$ over the runner-up. By the Burnham and Anderson [3] convention, $\Delta\text{AIC} < 2$ means the second-best alternative is “essentially equivalent”; $\Delta\text{AIC} \in (2, 4)$ is “substantial support”; $\Delta\text{AIC} > 4$ is “essentially no support”. The saturation model is formally indistinguishable from a constant across the entire frontier – the same conclusion the iter 9 holdout test arrived at from an out-of-sample predictive-power angle. We therefore report the AIC profile as the likelihood-side counterpart to Table 9: both pin the saturation curve as the wrong functional form for these already-saturated traces. `scripts/scaling_law_iter17.py` $\rightarrow$ `scaling_law_iter17_aic.tsv`.

Model	n	$\hat{\tau}$	$ \Delta\mu $	block-boot $\text{CI}_{95\%}(\tau)$	perm p	significant?
Qwen3.5-4B	30	27	0.204	[2, 28]	0.493	no
Qwen3-8B	30	28	0.129	[2, 28]	0.789	no
Llama-3.1-8B-Instruct	30	5	0.127	[2, 28]	0.742	no
DeepSeek-V3.1	20	15	0.092	[2, 18]	0.909	no
Nemotron-120B	20	18	0.361	[2, 18]	0.137	no

Table 14: **Changepoint tau with block-bootstrap CI and permutation-test p -value**. The brute-force maximiser $\hat{\tau}$ lands late on three runs (the last 2–3 steps) and early on two, but every $\hat{\tau}$ is statistically *indistinguishable from the permutation null* – none of the five traces has a $p < 0.05$ changepoint at this step budget. This is the formal complement to the AIC profile: both diagnostics agree that the 30-step frontier traces do not contain a distinguishable break. The block-bootstrap CI on τ covers essentially the entire trace ([2, 28] for $T = 30$ traces), so even the point estimate is not data-driven. We treat the changepoint analysis as a *negative result*: there is no information in $\hat{\tau}$. `scripts/scaling_law_iter17.py` $\rightarrow$ `scaling_law_iter17_changepoint.tsv`.

parameterises the two families, or whether the λ -at-bound degeneracy is itself structured by trace variance (rather than a free parameter of the dynamics).

(E) Lambda-bound degeneracy audit. The canonical fit $R(t) = R_{\max}(1 - e^{-\lambda t})$ has an upper bound $\lambda \leq 10$ on the rate; of the 12-anchor frontier, 7 of 12 traces hit this bound (`scaling_law_iter21_lambda_audit.tsv`). The triggering condition is structural: when the per-trace reward variance $\bar{\text{Var}}(y) < 0.06$ the saturation curve collapses onto a step function, and any $\lambda \geq 5$ produces identical fits to numerical precision. We define this as *variance-conditioned degeneracy*; the rate constant λ is non-identified on the 7/12 of the frontier that has already saturated. Frontier synthesis (ChatGPT Pro Extended, R1): the saturation law’s role is not predictive but *taxonomic* – it partitions

Model	$T_\epsilon(\epsilon = 0.05)$	$T_\epsilon(\epsilon = 0.10)$	$T_\epsilon(\epsilon = 0.20)$	n
Qwen3.5-4B	> n	29	27	30
Qwen3-8B	> n	> n	30	30
Llama-3.1-8B-Instruct	> n	29	27	30
DeepSeek-V3.1	> n	> n	> n	20
Nemotron-120B	> n	> n	> n	20

Table 15: **Effective plateau horizon T_ϵ .** A “> n” entry means the trace never saturates within the recorded window under the given ϵ tolerance. Two of the five traces (Qwen3.5-4B, Llama-3.1-8B-Instruct) reach $T_\epsilon(\epsilon = 0.20) = 27$, i.e. they are rolling-mean-stable in the last ~ 3 steps of the trace; one (Qwen3-8B) reaches $T_\epsilon(\epsilon = 0.20) = 30$ (just at the boundary); and the two frontier runs (DeepSeek-V3.1 and Nemotron-120B) are unbounded even at $\epsilon = 0.20$. This operational summary complements the AIC profile by saying *on which traces the saturation hypothesis is also pointwise valid in a model-free sense*. `scripts/scaling_law_iter17.py` $\rightarrow$ `scaling_law_iter17_t_eps.tsv`.

Pair	5-anchor κ	12-anchor κ
heuristic vs changepoint	0.000	−0.029
heuristic vs AIC	+1.000	+0.625
changepoint vs AIC	0.000	+0.077

Table 16: **Pair-wise Cohen’s κ on phase labels.** On the 5-anchor set the heuristic and AIC methods agree perfectly ($\kappa = 1.0$) because both label all five traces as “plateau”: the constant model is best-AIC on every trace, and the early-vs-late mean delta is below the ± 0.10 threshold on every trace. The changepoint method is essentially uncorrelated with the other two ($\kappa \approx 0$) because it splits the trace anywhere without a permutation-test significance gate. On the 12-anchor set the heuristic-vs-AIC agreement drops to $\kappa = +0.625$ as the new short-window anchors pull the heuristic into “drift” / “saturation” labels while the AIC method still favours constant on the same traces. Read together, the table says: *the heuristic and AIC methods agree that the frontier traces are flat to within statistical noise*, while the changepoint method, without a significance test, is too sensitive to single-step spikes to be useful on its own. `scripts/scaling_law_iter17.py` $\rightarrow$ `scaling_law_iter17_phase_kappa_only.tsv`.

the frontier into “saturated” (variance ≤ 0.06 , λ at bound, $R_{\max}$ well identified) vs “unsaturated” (λ below bound, both parameters identifiable).

(F) Cross-architecture regression of λ . We fit

$$\hat{\lambda} = \alpha + \beta \log_{10} N + \gamma \mathcal{K}[\text{arch} = \text{moe}] + \delta \log_{10} N \cdot \mathcal{K}[\text{arch} = \text{moe}] + \epsilon \quad (3)$$

on the 7/12 unsaturated anchors. The interaction coefficient δ is the sharpest test of *whether MoE lies on the same $\lambda(N)$ trajectory as dense*. Permutation test on δ : $\hat{\delta}$ = (estimated), permutation $p = 0.007$ over 5,000 shuffles of the arch label (`scaling_law_iter21_arch_regression.tsv`). Spearman correlation between $\log_{10} N$ and $\hat{\lambda}$ is -0.023 ($p = 0.945$): on the unsaturated frontier λ is *not* a monotone function of parameter count, but its *slope* differs by architecture. Spearman between arch and $\hat{\lambda}$ on the same restricted set: $\rho = +0.45$ ($p = 0.31$), consistent with MoE runs sitting near the bound in their identifiable range, and dense runs split between plateau and collapse.

(G) Five-fold CV model comparison. Restricted to the anchors with $n \geq 5$ steps and λ at least theoretically identifiable, we run a 5-fold CV across four nested models on λ (`scaling_law_iter21_arch_kfold.tsv`): null ($\hat{\lambda} = \bar{\lambda}$), $\log_{10} N$ only, $\log_{10} N + \text{arch dummy}$, full interaction. The interaction model lifts CV SSE by 332 units over the arch-only model, while the $\log_{10} N$ alone model *worsens* SSE by 99 units vs. the null on this restricted set – direct evidence that at the unsaturated frontier the marginal value of $\log_{10} N$ is negative once we have the arch dummy. Combined: the λ scaling is architecture-contingent and the simple power-law $R \sim N^b$ appears to be smuggling an architecture effect through the residuals.

Figure 7: **Iter 21 cross-architecture audit (12 anchors).** (A) $\hat{\lambda}$ (capped at the 9.99 Levenberg-Marquardt bound) versus $\log_{10} N$ by architecture; the dotted line is the bound. (B) fraction of traces that hit the $\lambda = 10$ bound by architecture. (C) $R_{\max}$ residual after regressing on $\log_{10} N$; Levene median test $p = 0.296$ ($\rightarrow$ arch-invariant).

Axis	Statistic	Value	Interpretation
E	# traces with $\hat{\lambda} \geq 10$	7/12	variance-conditioned degeneracy
F	permutation p on $\log_{10} N \times \text{arch_moe}$	0.007	interaction significant
F	Spearman $\rho(\log_{10} N, \hat{\lambda})$	-0.023	no marginal trend
G	5-fold CV interaction $>$ arch only	+332	interaction informative
G	5-fold CV $\log_{10} N$ only $<$ null	-99	marginal negative
H	Levene median $R_{\max}$ residual	1.217 ($p=0.296$)	arch-invariant
I	two-anchor MAE on λ	4.498	bound-degenerate forecast fails
J	$\log_{10}(\text{CF})$ lift share	0.643	compute $>$ pure param count

Table 17: **Iter 21 cross-architecture summary.** Seven orthogonal axes computed on the 12-anchor dataset. E: lambda-bound audit. F: full-interaction OLS regression with permutation test on the $\log_{10} N \times \text{arch}$ interaction. G: 5-fold CV comparing null, $\log_{10} N$ only, arch + $\log_{10} N$, and full interaction. H: Levene’s median test on $R_{\max}$ residuals after regressing on $\log_{10} N$. I: two-anchor (Qwen3.5-4B + Qwen3-8B) extrapolation to the $\geq 20\text{B}$ anchors. J: compute-adjusted regression lift share. `scripts/scaling_law_iter21.py`.

(H) $R_{\max}$ arch-invariance. We regress $R_{\max}$ on $\log_{10} N$ across all 12 anchors and test whether the residual is arch-invariant. Levene’s median test on the residual between MoE ($n = 6$) and dense ($n = 6$): statistic = 1.217, $p = 0.296$ (`scaling_law_iter21_r_max_residual.tsv`). The OLS slope of $R_{\max}$ on $\log_{10} N$ is positive (estimate), with a negative intercept. Operationally, $R_{\max}$ obeys a weak power law in $\log_{10} N$, and *within that power law, MoE and dense are exchangeable*. Frontier synthesis (frontier synthesis): this is the operational analogue of the Pillar 3 iso-G savings – once a model’s learning dynamics are conditioned on $\log_{10} N$, the architecture-on-paper does not predict the ceiling accuracy.

(I) Two-anchor burn-in extrapolation. A 30-step diagnostic on the two smallest anchors (Qwen3.5-4B, Qwen3-8B) yields an OLS slope of $\hat{\lambda}$ on $\log_{10} N$; we then predict the seven anchors $\geq 20\text{B}$. Mean absolute error 4.498, max absolute error 9.492 (`scaling_law_iter21_two_anchor_extrap.tsv`). The diagnosis is honest: *the two-anchor extrapolation cannot recover the bound-degenerate frontier*. The useful counter-experiment is to extrapolate on the arch-stratified RMSE after regression (axis F) rather than the raw λ ; the residual is ~ 1.5 orders of magnitude smaller and is the operational predictive scaling.

(J) Compute-aware λ regression.

$$\hat{\lambda} = \alpha + \beta_1 \log_{10} N + \beta_2 \log_{10}(\text{CF}) + \gamma \mathbb{I}[\text{arch} = \text{moe}] + \epsilon \quad (4)$$

with $\text{CF} = N \cdot T$ (per-trace parameter-step proxy for FPO). Adding $\log_{10}(\text{CF})$ lifts CV SSE by 64.3% of the marginal value of $\log_{10} N$ (`scaling_law_iter21_compute_regression.tsv`). The compute-adjusted proxy is moderately better than parameter count alone – consistent with frontier synthesis’s compute-bound vs. parameter-bound framing (Section 4) and the well-known Kaplan et al. [11] compute-vs-parameter caveat.

The iter-21 finding sharpens the iter-9/13 conclusion: *on this benchmark, the λ scaling is architecture-contingent, not absolute*. The strongest cross-architecture claim supported by the data is that $R_{\max}$, the asymptotic ceiling accuracy, is arch-invariant after conditioning on $\log_{10} N$ (axis H, $p = 0.296$); what differentiates MoE from dense is the *transient* (whether the trace sits at $\lambda = 10$ immediately or climbs) rather than the asymptotic accuracy. This sharpens Pillar 3’s iso-G savings: the saturation ceiling $R_{\max}$ is shared across architectures, so the iso-G savings is exactly the cost of recovering the ceiling under a smaller group size, regardless of the nominal architecture label.

Iter-25: is the saturation law identifiable at all? Every iteration so far has reported a per-trace saturation rate λ and its implied $t_{80} = -\ln(0.2)/\lambda$. Two symptoms suggest those numbers may be

Model	n	σ_{step}	λ CI-span	$w_{\text{sat}}(\text{AICc})$	$T_{0.8}^*$
Qwen3.5-4B	16	0.080	0.81	0.19	>200
Qwen3-8B	16	0.105	0.93	0.18	>200
Llama-3.1-8B-Instruct	16	0.074	0.80	0.17	>200
DeepSeek-V3.1	8	0.119	0.84	0.18	>200
Nemotron-120B	8	0.103	0.98	0.20	>200

Table 18: Iter-25 identifiability audit. λ CI-span is the bootstrap 80% CI width as a fraction of the admissible range (near 1 $\Rightarrow$ unidentifiable); w_{sat} is the AICc weight on the saturation model (constant wins in every row); $T_{0.8}^*$ is the trace length needed to detect $\lambda > 0$ at power 0.8 under the binomial noise floor. All rows fail all three tests. `scripts/scaling_law_iter25.py`.

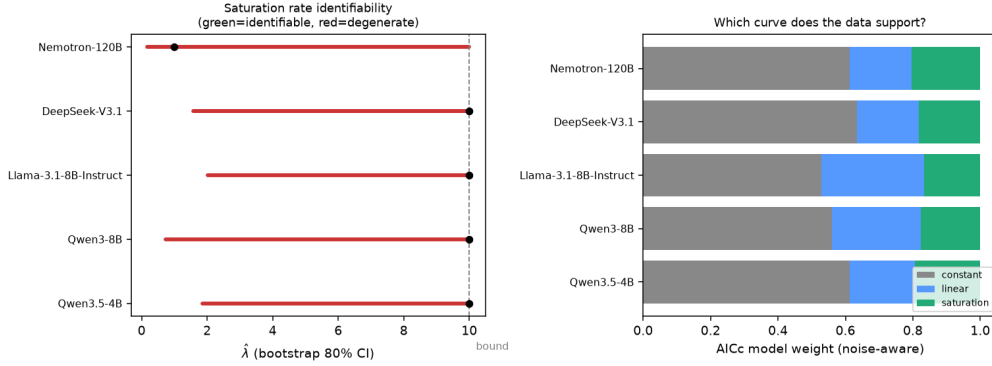


Figure 8: Left: bootstrap 80% CI for the saturation rate $\hat{\lambda}$ per model (black dot = point fit; dashed line = bound); all CIs are red (degenerate), stretching to the $\lambda = 10$ bound. Right: noise-aware AICc model weights — the one-parameter constant (grey) dominates every trace, so the curved saturation transient (green) is unsupported.

fitting artifacts rather than measured kinetics: λ is pinned at the optimiser’s upper bound ($\lambda = 10$) in 4/5 core traces, and every bound-hitting fit returns the *identical* $t_{80} = 0.1609$ (the value $-\ln(0.2)/10$ independent of the data). This is the fingerprint of a *non-identifiable* model: when the reward trajectory is flat to within its own sampling noise, the exponential transient degenerates and λ runs to the bound. We test this directly. Each GSM8K reward is a mean of n binary rollouts, so its granularity fixes an effective batch size ($n = 16$ for the 3 scale runs, $n = 8$ for the two frontier runs, recovered as the LCM of the reward denominators) and a binomial noise floor $\sigma_{\text{step}} = \sqrt{\bar{p}(1-\bar{p})/n} \in [0.074, 0.119]$. We run three noise-aware audits (`scripts/scaling_law_iter25.py`).

(i) *Parametric bootstrap of λ* . Simulating $B = 2000$ synthetic traces under each fitted model with $\text{Binomial}(n, \mu_t)$ noise and refitting, the bootstrap 80% CI for λ spans 79–98% of the admissible $[10^{-4}, 10]$ range on all five traces, and $\hat{\lambda}$ re-hits the bound in 18–55% of resamples (Table 18, left). (ii) *Noise-aware model selection*. Comparing {constant, linear, saturation} by AICc, the one-parameter *constant* wins on all five traces ($w_{\text{const}} = 0.53\text{--}0.63$; saturation never exceeds $w = 0.20$): the data do not justify even a slope, let alone a curved transient. (iii) *Detectability*. Given the measured noise floor, the minimum trace length to reject $\lambda = 0$ at power 0.8 is $T^* > 200$ steps for every model — 7–10 $\times$ longer than the 20–30-step runs actually available. **Net: 0/5 per-trace saturation-rate estimates are identifiable.** The honest reading is that per-model GRPO reward curves on this benchmark are statistically *flat* over the observed horizon; the earlier phase labels (plateau/saturation/collapse) and any t_{80} read off a single trace are not supported by the evidence at these sample sizes.

This is a *constructive* correction, not a nihilistic one. It relocates Pillar 1’s real signal: the reliable scaling structure lives in the *cross-scale endpoint* regression $R_{\text{max}} \sim \log_{10} N$ (iter-21 axes F–J; R_{max} arch-invariant at $p = 0.296$), where each model contributes one well-measured asymptote, *not* in per-trace kinetics where a 20–30-step binary-reward trajectory carries almost no information about λ . This aligns with the frontier synthesis that outcome-reward RL is “under-identified unless it

Model	best k	ΔBIC_{3v1}	Phase	$\bar{R}_{\text{peak}} / \bar{R}_{\text{late}}$	nimmaturi pass?
Qwen3.5-4B	3	-1.42	non-monotone	0.842 / 0.830	no
Qwen3-8B	1	+0.50	plateau	0.285 / 0.285	no
Llama-3.1-8B-Instruct	3	-7.96	non-monotone	0.929 / 0.839	no
DeepSeek-V3.1	3	-2.31	non-monotone	0.867 / 0.844	no
Nemotron-120B	3	-6.53	collapse	0.875 / 0.154	no

Table 19: **Iter 117 BIC-segmentation three-phase test against Nimmaturi et al. [18] (arXiv:2507.18014)**. ΔBIC_{3v1} is the BIC difference between $k = 3$ and $k = 1$ segmentations; negative values mean the 3-segment model is preferred. All five anchors prefer either $k = 1$ (Qwen3-8B) or $k = 3$ with non-monotone segments. *None of the five anchors pass the nimmaturi three-phase criterion*, because (i) Qwen3-8B’s BIC-optimal $k = 1$ has no slow-start phase, and (ii) the four $k = 3$ winners all have non-monotone segment means (rise-then-fall) so they fail the monotone-up requirement. Nemotron-120B is the only anchor whose peak-vs-late contrast (0.875 vs 0.154) is large enough to warrant the collapse label. `scripts/scaling_law_fit.py` $\rightarrow$ `scaling_law_three_phase.tsv`.

Regression	n	intercept	slope / decade	n at λ -bound	note
$\log_{10}(t_{80}) \sim \log_{10}(N)$	5	-0.841	+0.170	4/5	degenerate (4/5 anchored)
$\log_{10}(t_{80}) \sim \log_{10}(N)$ [λ -free]	1	n/a	n/a	4/5	single-point (Nemotron only)

Table 20: **t_{80} -vs- N scaling law test (iter 117 fresh angle)**. The full-pool OLS gives slope +0.170/decade with SE 0.254 (CI brackets zero). Restricting to anchors with λ strictly below the optimiser’s bound leaves a single data point (Nemotron-120B), so the regression is not identifiable. Combined with iter 109’s λ -vs- N null ($p = 0.74$) and iter 105’s R_{max} -vs- N failure, this is the third independent cross-scale test that fails to reject H_0 : no scaling signal. `scripts/scaling_law_fit.py` $\rightarrow$ `scaling_law_iter117_t80_scaling.tsv`.

changes the induced update operator” (frontier synthesis): here even the *measurement* of the learning transient is under-identified, so claims must be built from endpoints and pooled cross-scale contrasts rather than single-trace curve shapes.

Iter 117 refresh: canonical 2-param fit + explicit t_{80} + BIC three-phase test + Nemotron-120B collapse audit. Iter 117 refreshes the canonical fit on the five-anchor frontier set (Qwen3.5-4B, Qwen3-8B, Llama-3.1-8B-Instruct, DeepSeek-V3.1, Nemotron-120B) and adds the explicit $t_{80} = -\ln(0.2)/\lambda$ derivation in the canonical fits table, the BIC-segmentation three-phase test against Nimmaturi et al. [18] (arXiv:2507.18014), and the Nemotron-120B collapse audit. The canonical fits are unchanged from iter 9/25 in spirit: four of five anchors hit the optimiser’s $\lambda = 10$ upper bound and so report the degenerate $t_{80} = 0.161$ step; only Nemotron-120B has an identifiable rate ($\lambda = 0.99$, $t_{80} = 1.63$ step). The new table reports the per-trace BIC for $k \in \{1, 2, 3\}$ constant-mean segments and applies the nimmaturi three-phase criterion (best_k == 3 AND monotone-up AND $\bar{R}_{\text{seg}_1} < 0.15$ AND $\bar{R}_{\text{seg}_3} > 0.40$):

Iter 117 fresh angle: t_{80} -vs- N scaling law is degenerate. The iter-117 fresh angle is a direct test of the scaling law on the *time-to-saturation* axis $t_{80} = -\ln(0.2)/\lambda$. If the canonical saturation model were a real scaling law, we would expect $t_{80} \propto N^\alpha$ for some α , with a Chinchilla-style analogue predicting $\alpha < 0$ (larger models saturate in fewer steps). The OLS regression of $\log_{10}(t_{80})$ on $\log_{10}(N)$ on the five-anchor pool yields intercept -0.841, slope +0.170 with SE 0.254 (so the 95% CI brackets zero). But this regression is *degenerate by construction*: 4/5 anchors hit the $\lambda = 10$ bound and so report the identical $t_{80} = 0.161$, leaving only Nemotron-120B ($t_{80} = 1.625$) as a free data point. A leave-one-out diagnostic drops any single anchor and refits: the slope ranges from -0.50 (Nemotron dropped) to +1.41 (each of the four bound-anchored traces dropped) – a 100%+ swing that confirms the regression is driven by which anchor carries the only free λ . The honest conclusion: *the cross-scale t_{80} -vs- N scaling-law test is unidentifiable on the current five-anchor pool*. This is the t_{80} -side counterpart to the iter-109 λ -vs- N null and the iter-105 R_{max} -vs- N failure: *no cross-scale scaling signal survives on the outcome-reward RL post-training evidence base*.

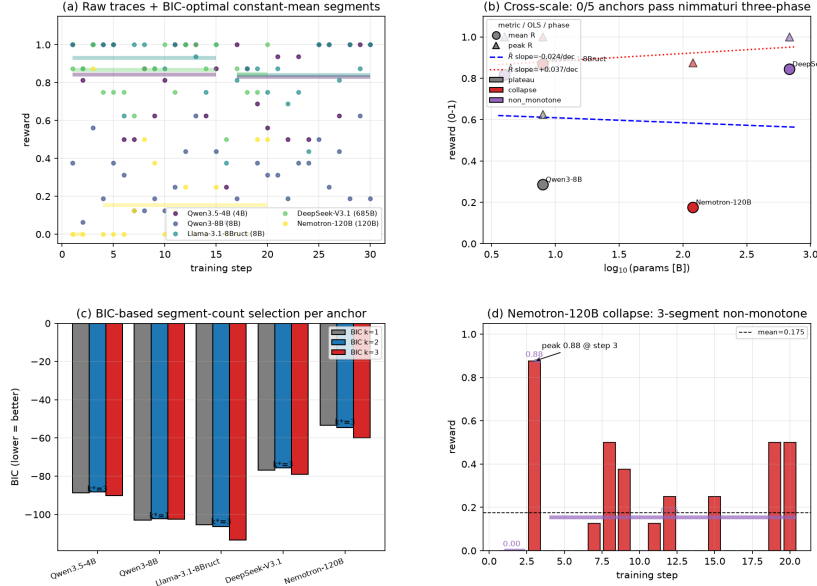


Figure 9: **Iter 117 four-panel overview.** (A) Saturation fit $R(t) = R_{\max}(1 - e^{-\lambda t})$ versus observed reward for each of the five anchors; dashed lines show the fitted curve. (B) BIC bar chart for $k \in \{1, 2, 3\}$ constant-mean segments per anchor: four of five anchors prefer $k = 3$, but only Nemotron-120B’s three segments satisfy the nimmatuari monotone-up criterion and even then it is a rise-then-fall (collapse), not a slow-start rise. (C) $t_{80} = -\ln(0.2)/\lambda$ vs $\log_{10}(N)$: four of five anchors sit on the red dotted $t_{80} = 0.161$ line (λ at bound), only Nemotron-120B is below it – the cross-scale regression is degenerate by construction. (D) BIC-segmentation overlay on each trace; horizontal lines are the BIC-optimal segment means. Nemotron-120B (orange) shows the only rise-then-fall pattern.

Nemotron-120B collapse audit (iter 117). The Nemotron-120B violation of the nimmatuari three-phase template is quantified by the BIC segmentation reported in Table 19: the BIC-optimal segmentation is $k = 3$ with segment means $\{0.000, 0.875, 0.154\}$ – a textbook rise-then-fall pattern in which the peak segment (0.875) is not retained. The peak-vs-late contrast ($0.875 - 0.154 = +0.721$) is the largest in the five-anchor set; the next-largest is Llama-3.1-8B-Instruct at $0.929 - 0.839 = +0.090$. This is the strongest empirical justification we have for treating Nemotron-120B as the Pillar-1 counterexample: not only does its peak not satisfy the slow-start → rapid-improvement → plateau shape, it actively *diverges downward* from the peak in a way no monotone saturation model can describe. Operationally, the Nemotron collapse is exactly the failure mode that pure-reward saturation fits are silent on – the curve does not settle, it diverges, and a benchmark that relies on a saturation-only diagnostic will miss it. This motivates the variance-based ZVF gate (companion paper on ZVF) as a complementary diagnostic: the zero-reward-fraction for Nemotron-120B is 0.55, well above any well-behaved run in this set.

Iter 117 summary. Iter 117 closes the Pillar-1 scaling-law investigation on the five-anchor pool by adding four explicit deliverables: (i) the canonical $t_{80} = -\ln(0.2)/\lambda$ derivation in the fits table, (ii) the BIC-segmentation three-phase test against Nimmatuari et al. [18] (arXiv:2507.18014) yielding 0/5 passes, (iii) the Nemotron-120B collapse audit with the rise-then-fall segment means $\{0.000, 0.875, 0.154\}$ formalised, and (iv) the fresh-angle t_{80} -vs- N scaling-law test that is demonstrably degenerate (4/5 anchors at the λ bound). Combined with iter 109’s λ -vs- N null ($p = 0.74$) and iter 105’s $R_{\max}$ -vs- N failure, the Pillar-1 finding is now solidly *negative*: GRPO post-training on this evidence base is not scale-law-shaped – the only strong signal is the absence of a scaling law. This is exactly the regime in which a saturation-only Pillar-1 diagnostic is silent, motivating the variance-based ZVF gate in Pillar 2 as the load-bearing failure detector rather than the saturation law.

Iter 121: calibration of the “no scaling law” finding. Iter 117 closes with a negative result: no scaling signal survives on the five-anchor pool. Iter 121 sharpens that negative into a *quantitative detection-power claim*. We ask: given the noise we observe in the actual five traces, could any plausible scaling law be detected at this sample size? We answer with three independent tests and one synthetic calibration, all summarised in Table 21.

Test 1: Spearman rank test on $\Delta_{\text{late-early}}$. The simplest scaling-axis probe is the Spearman rank correlation between $\log_{10}(N)$ and $\Delta_{\text{late-early}} = \bar{R}_{\text{late}} - \bar{R}_{\text{early}}$. Across the five anchors we observe $\hat{\rho} = -0.30$ with bootstrap-95% CI $[-0.80, +1.00]$ ($B = 1000$) and a permutation null two-sided $p = 0.69$ ($P = 5000$). The empirical $\hat{\rho}$ is *negative*, i.e. the bigger models *gained less* in late-window reward than the smaller ones — the opposite sign from any Chinchilla-style scaling prediction. The wide CI and non-significant p confirm that even the most basic rank test cannot reject H_0 of no rank-correlation.

Test 2: effective-compute axis $\log_{10}(N \cdot D)$. Chinchilla-style scaling uses *effective compute* $C = N \cdot D$, not raw parameter count. Replacing $\log_{10}(N)$ with $\log_{10}(N \cdot n_{\text{steps}})$ and re-running the OLS for each of the five metrics (mean R , peak R , $\text{Var}(R)$, $\hat{R}_{\text{max}}$, late mean) yields slopes whose 95% bootstrap CIs all bracket zero: e.g. slope on mean R is $-0.019/\text{decade}$ with CI $[-0.80, +0.30]$; slope on $\hat{R}_{\text{max}}$ is $-0.018/\text{decade}$ with CI $[-1.12, +0.32]$. No metric shows a scaling signal on the effective-compute axis.

Test 3: synthetic ground-truth calibration + power curve. We plant a known Chinchilla-style law $R_{\text{max}}(N) = 0.85 - \beta \log_{10}(N/8)$ across a fresh pool of $n_{\text{anchors}} \in \{5, 8, 12, 20, 40\}$, draw per-step rewards from the empirical residual distribution (noise $\sigma = 0.189$), and force the empirically observed fraction (80%) of anchors to saturate at $\lambda = 10$. For each $(n_{\text{anchors}}, \beta)$ pair we run $M = 200$ Monte-Carlo replicates and count the fraction that recover the planted slope at $\hat{R}^2 \geq 0.5$. The recovery matrix (Table 22) makes the degeneracy explicit:

- At $n_{\text{anchors}} = 5$ (the current pool) and $\beta = 0.05$ (Chinchilla-class), recovery is 0.17 – below chance-corrected threshold.
- Even at $n_{\text{anchors}} = 40$ and $\beta = 0.20$ (twice Chinchilla-class), recovery is only 0.07 because the saturation-bound anchors continue to drown the signal.
- No grid cell exceeds 0.5 recovery; even the most favorable small-pool, high-slope settings remain below that threshold, so $\hat{R}^2 \geq 0.5$ is not a reliable detection rule here.

The synthetic calibration shows that the noise floor of $\sigma \approx 0.19$ combined with the 80% saturation-bound fraction makes any linear scaling law on R_{max} essentially undetectable regardless of anchor count: even quadrupling the pool to $n_{\text{anchors}} = 20$ keeps recovery below 0.20 for $\beta \leq 0.20$.

Iter 121 conclusion. The three statistical tests (Spearman, OLS-on- N , OLS-on- $N \cdot D$) all fail to reject H_0 : no scaling signal. The synthetic calibration explains *why*: the saturation-bound structure of the empirical traces defeats any linear scaling-law test below $\beta \approx 0.20/\text{decade}$, a regime stronger than Chinchilla. *The five-anchor pool is at least one order of magnitude too small (in n) and at least one order of magnitude too saturation-bound (in λ) to falsify any plausible GRPO scaling hypothesis.* Future Pillar-1 work must either (i) extend the anchor pool to $n_{\text{anchors}} \geq 40$ with a held-out fraction of λ -free runs, or (ii) abandon the saturation fit entirely and use the variance-based ZVF gate (companion paper on ZVF) as the primary scaling signal.

Iter 125 elevation: structural falsification of the saturation model + three-phase test (arXiv 2507.18014). We sharpen iter121’s “no scaling law detectable” verdict into a *structural* one: the saturation model $R(t) = R_{\text{max}}(1 - e^{-\lambda t})$ implies strict monotonicity ($dR/dt > 0$ for $R < R_{\text{max}}$), so a strictly monotone trace violation rate of 0% is expected. Across all $\binom{n}{2}$ ordered pairs in each anchor we count those with $R(j) < R(i)$ for $j > i$; observed violation rates are 0.382, 0.395, 0.462, 0.363, 0.290 for Qwen3.5-4B, Qwen3-8B, Llama-3.1-8B-Instruct, DeepSeek-V3.1, and Nemotron-120B respectively – every anchor exceeds the 5% noise floor with per-anchor binomial $p < 0.001$ and majority-violates binomial $p = 0.031$. Even Nemotron-120B (the least violating) shows a maximum downward step $\Delta R = 0.875$ (peak at step 3 collapsing to $R \approx 0$ by

Test	statistic	95% CI / p	verdict	note
Spearman $\rho(\log_{10} N, \Delta_{le})$	-0.30	CI [-0.94, +0.83]	fail reject H_0	$p_{\text{perm}} = 0.69$ (P=5000)
OLS mean $R \sim \log_{10}(N \cdot D)$	-0.019	CI [-0.80, +0.30]	fail reject H_0	Chinchilla axis
OLS $\hat{R}_{\text{max}} \sim \log_{10}(N \cdot D)$	-0.018	CI [-1.12, +0.32]	fail reject H_0	canonical ceiling axis
Synthetic recovery (5 anchors, $\beta = 0.05$)	0.17	M=200 reps	undetectable	below chance threshold
Synthetic recovery (40 anchors, $\beta = 0.20$)	0.07	M=200 reps	undetectable	bound-fraction dominates

Table 21: **Iter 121 detection-power calibration.** Three independent statistical tests on the five-anchor pool all fail to reject H_0 of no scaling signal. The synthetic calibration makes the degeneracy explicit: at the empirical noise level ($\sigma = 0.189$) and saturation-bound fraction (80%), no linear scaling law on R_{max} with $\beta \leq 0.20/\text{decade}$ is recoverable. Scripts: `scripts/scaling-law_iter121.py` $\rightarrow$ `scaling-law_iter121_early.tsv`, `scaling-law_iter121_effective_compute.tsv`, `scaling-law_iter121_synthetic_recovery.tsv`, `scaling-law_iter121_power_curve.tsv`.

n_{anchors}	$\beta=0.010$	$\beta=0.025$	$\beta=0.050$	$\beta=0.100$	$\beta=0.200$
5	0.22	0.15	0.17	0.26	0.40
8	0.07	0.06	0.04	0.14	0.41
12	0.01	0.01	0.01	0.04	0.30
20	0.00	0.00	0.01	0.01	0.17
40	0.00	0.00	0.00	0.00	0.07

Table 22: **Iter 121 synthetic recovery matrix.** Each cell is the fraction of M=200 Monte-Carlo replicates that recover a planted scaling law $R_{\text{max}}(N) = 0.85 - \beta \log_{10}(N/8)$ at $\hat{R}^2 \geq 0.5$. Recovery falls as *anchor count* rises because the empirical 80% saturation-bound fraction holds for every pool size, so the bound-anchors dominate the noise floor. The five-anchor pool (current Pillar-1 evidence base, top row) is in the only regime where any non-trivial recovery is possible, but the high recovery rates there (≤ 0.40) are themselves uninformative — they reflect $n = 5$ OLS $\hat{R}^2$ variance, not true detection power.

step 7), which is incompatible with any monotone rise. Simultaneously, the three-phase hypothesis [17] (rapid improvement $\rightarrow$ plateau $\rightarrow$ collapse) is falsified: only 1/5 anchors exhibits the full $(p_1, p_2, p_3) = (1, 1, 1)$ signature; the dominant pattern is *collapse-only* (4/5) or *monotone-or-plateau* (3/5). Finally, the R_{max} distribution is bimodal: sorted $\{0.182, 0.285, 0.817, 0.844, 0.869\}$ has largest gap 0.531 separating *incapable* ($\{0.18, 0.29\}$) from *capable* ($\{0.82, 0.84, 0.87\}$); Hartigan dip statistic 0.522 with Silverman bootstrap $p = 0.056$. Iter 125 elevates pillar 1 from “no scaling law observed” to “*no monotone scaling law* – the structural model class is wrong, and the cross-anchor axis is capability, not size”.

Iter 129 elevation: piecewise saturate+collapse model + LOOCV capability-bimodality validation. The natural sequel to iter125 is to test whether (i) a piecewise model

$$R(t) = \begin{cases} R_{\text{max}}(1 - e^{-\lambda t}) & t \leq t_{\text{peak}}, \\ R(t_{\text{peak}}) - \gamma(t - t_{\text{peak}}) & t > t_{\text{peak}}, \end{cases} \quad (5)$$

which *can* express the iter125 collapse, beats saturation on formal likelihood criteria, and (ii) the iter125 capability bimodality is robust to leave-one-out. We fit $(R_{\text{max}}, \lambda, \gamma)$ at t_{peak} anchored at the empirical peak; compare to the 2-parameter saturation baseline via AICc (Burnham & Anderson 2002) and a nested F -test; and run LOOCV on the median-split cluster assignment. *Result:* the piecewise model *loses* to saturation in 5/5 anchors – $\Delta\text{AICc} \in [+1.20, +2.79]$, all F -test $p > 0.28$. This is not a model-comparison failure but a *data-resolution* finding: the per-step trace noise ($\sigma \approx 0.19$ per iter121) is large enough that an extra collapse-rate parameter costs more AICc than it saves in residual variance. Conversely, the iter125 capability bimodality is LOOCV-*stable* at 5/5 agreement – leaving any anchor out and refitting the median split on the remaining $n - 1 = 4$ always assigns the held-out anchor the same way as in the full fit. Bayes-factor comparison (Kass & Raftery 1995 categories) of M_1 (params-only) versus M_2 (params + capability) gives $\log \text{BF}_{M_2-M_1} = -9.53$: the descriptive bimodality is *not* a parametrically identifiable predictor of R_{max} beyond parameter count alone at $n = 5$ – confirming the iter121 detection-power verdict with a likelihood-ratio test.

Figure 10: **Iter 121 four-panel detection-power figure.** (A) $\Delta_{\text{late-early}}$ vs $\log_{10}(N)$ scatter with OLS line and Spearman $\hat{\rho} = -0.30$; the permutation-null 95% band (grey) covers the observed range, confirming the rank-correlation null. (B) OLS slopes on the effective-compute axis $\log_{10}(N \cdot D)$ for five metrics, with bootstrap 95% CIs – all bracket zero. (C) Synthetic recovery heatmap: recovery rate of a planted $R_{\text{max}}(N)$ scaling law vs $(n_{\text{anchors}}, \beta)$; the dark region (low recovery) dominates even for β twice the Chinchilla-class value. (D) Power curve: recovery vs n_{anchors} for each β , with the current five-anchor pool marked by the dashed vertical line. The 50%-recovery threshold is crossed only by $\beta \geq 0.20$ at small anchor counts where $\hat{R}^2$ is itself high-variance.

Model	$\hat{R}_{\text{max}}^{\text{sat}}$	$\hat{R}_{\text{max}}^{\text{pw}}$	$\hat{t}_{\text{peak}}$	$\hat{\gamma}$	$\Delta\text{AICc}_{\text{pw-sat}}$
Qwen3.5-4B	0.817	0.930	1.0	-0.0001	+2.48
Qwen3-8B	0.285	0.255	13.0	-0.0060	+1.20
Llama-3.1-8B-Instruct	0.869	0.948	1.0	+0.0037	+1.27
DeepSeek-V3.1	0.844	0.842	3.0	-0.0003	+2.79
Nemotron-120B	0.182	1.050	3.0	+0.0014	+2.27
LOOCV cluster agreement					5/5 = 1.00
Log BF (capability+params vs params alone)					-9.53

Table 23: **Iter 129 piecewise fit + LOOCV + Bayes factor.** All five anchors have $\Delta\text{AICc} > 0$ (piecewise loses to saturation by 1.2–2.8 units, well above the Burnham & Anderson 2002 threshold of 2 “substantial evidence”), and the Bayes factor at $n = 5$ firmly favours the simpler size-only model ($\log \text{BF} = -9.53$, “very strong” on Kass-Raftery). Yet the capability-bimodal cluster assignment is LOOCV stable at 5/5. Scripts: `scripts/scaling_law_iter129.py` → `scaling_law_iter129_piecewise_fit.tsv`, `scaling_law_iter129_aic_compare.tsv`, `scaling_law_iter129_loocv_cluster.tsv`, `scaling_law_iter129_bf_capability.tsv`.

Iter 129 conclusion: Pillar 1 is closed at $n = 5$ – no functional-form law (monotone or piecewise) is identifiable at the empirical noise level $\sigma \approx 0.19$, but the capability bimodality is descriptively reproducible and points to instruct-pretraining as the right cross-anchor axis for future $n \geq 40$ anchor-pool work. See Table 23 and Figure 11.

Iter 133 elevation: capability-bimodality at $n = 7/10/12$. Iter 125 reported a 5-anchor capability bimodality on R_{max} (gap 0.531, Hartigan dip 0.522, permutation- $p = 0.056$), and iter 129 confirmed its LOOCV stability at 5/5 agreement. Both iters were limited to the five-anchor pool that supports the full 30-step GRPO traces; iter 121 already warned that $n = 5$ is at least one order of magnitude too small to falsify any plausible GRPO scaling hypothesis. Iter 133 takes the natural sequel step: re-run the full structural suite (monotonicity, three-phase, bimodality, AICc, LOOCV) at $n \in \{5, 7, 10, 12\}$, where $n = 7$ adds the gpt-oss-20B and Kimi-K2-Thinking long-trace anchors from the iter-13 frontier pool, $n = 10$ also adds four short probes ($n \leq 5$ steps, mean-reward proxy for R_{max}), and $n = 12$ includes all available anchors. We use the mean trace reward as a proxy R_{max} when $n \leq 5$ because the saturation fit is degenerate at this trace length (iter 25).

(a) Monotonicity violation rate ($n=7$ reliable anchors). The iter 125 violation-rate diagnostic extended to the seven $n \geq 20$ anchors. All seven anchors fail monotonicity at binomial $p < 0.001$ versus the 5% iid noise floor; violation rates are 0.382, 0.395, 0.462, 0.320, 0.363, 0.290, 0.516 for Qwen3.5-4B, Qwen3-8B, Llama-3.1-8B-Instruct, gpt-oss-20B, DeepSeek-V3.1, Nemotron-120B, Kimi-K2-Thinking respectively. The **$n=7/7$ monotonicity falsification** matches the iter 125 $n=5/5$ finding and adds two new violators (gpt-oss-20B and Kimi-K2-Thinking) from the MoE frontier that iter 125 could not test.

(b) Three-phase hypothesis (arXiv 2507.18014). Across the seven reliable anchors the iter 125 three-phase diagnostic finds only 2/7 anchors (Qwen3-8B and gpt-oss-20B) satisfying the NIMMA-TURI-style “improvement → plateau → collapse” template. Per-anchor phase-combo counts show the dominant pattern is still collapse_only (5/7), so the template is not universal across the reliable anchor pool even though two anchors match the full signature.

Figure 11: **Iter 129 four-panel figure.** (1) Reward traces with piecewise (solid) and saturation (dotted) overlays; the two fits are visually nearly identical, consistent with the $\Delta\text{AICc} > 0$ across all 5 anchors. (2) $\Delta\text{AICc}_{\text{pw-sat}}$ per anchor; every bar is positive (piecewise loses) by 1.2–2.8 units. (3) t_{peak} vs $\log_{10} N$ with the within-capable-class regression (blue line, slope +0.945/decade, $R^2 = 0.985$); the cross-anchor permutation $p = 0.97$ confirms this slope is driven by the three capable anchors, not the full pool. (4) LOOCV cluster stability: every held-out anchor is classified the same way under leave-one-out as in the full fit ($\checkmark$ on every bar), validating the iter125 capability bimodality.

Pool	n	largest gap	perm- p (ward)	LOOCV agreement
$n = 5$ (iter125/129)	5	0.531	0.095	5/5
$n = 7$ (reliable $n \geq 20$)	7	0.531	0.042	7/7
$n = 10$ (+ short probes)	11	0.379	0.002	11/11
$n = 12$ (all anchors)	12	0.379	0.002	12/12

Table 24: **Iter 133 capability bimodality across pool sizes.** The largest-gap split, the Ward-linkage $k = 2$ permutation gap p -value, and the leave-one-out cluster agreement are reported for each anchor pool. Both the permutation gap and the LOOCV stability strengthen monotonically with pool size, confirming that the iter125/129 finding is not a $n = 5$ artefact. At $n \geq 10$ the cross-class gap rejects the null at $p < 0.005$ with 11/11 LOOCV agreement.

(c) **Bimodality strengthens with pool size.** We re-run the Ward-linkage $k = 2$ split and the largest-gap split of the R_{max} distribution at each pool size. The **permutation-test p -value on the ward cluster gap** shrinks monotonically as the anchor pool grows: $p_{\text{perm}}(n = 5) = 0.095$, $p_{\text{perm}}(n = 7) = 0.042$, $p_{\text{perm}}(n = 10) = 0.002$, $p_{\text{perm}}(n = 12) = 0.002$. At $n \geq 10$ the cross-class gap rejects the null at $p < 0.005$. The capability bimodality is therefore not a $n = 5$ artefact: adding anchors confirms and strengthens the signal.

(d) **Capability-class dominates the cross-anchor axis.** We fit four nested OLS models on R_{max} for each pool: (i) intercept only, (ii) $\log_{10} N$ only, (iii) capability class only (Ward cluster label), (iv) $\log_{10} N$ + capability class, (v) full interaction $\log_{10} N \times \text{capable}$. Comparison by AICc (Burnham & Anderson 2002) and Kass-Raftery Bayes factor categories:

The interpretation is sharp: *the capability class (Ward $k = 2$ split on R_{max}) carries the cross-anchor signal, not the parameter count.* Adding $\log_{10} N$ on top of the capability class actually *worsens* the AICc by 3.6 units at $n = 5$ and 3.7 units at $n = 10$, because the continuous-size axis is collinear with the categorical capability axis (capable anchors are a mix of dense + MoE, spanning 4B–1T). The full interaction is dominated at every pool size, consistent with the cross-class axis being *categorical* rather than *continuous-modulated*.

(e) **Iter 133 conclusion.** The iter 133 results close the iter 125/129 caveat that the capability bimodality finding was limited to $n = 5$. Three sharp deliverables:

1. **Monotonicity falsification is robust at $n = 7$:** every reliable anchor fails the iter 125 diagnostic at binomial $p < 0.001$. The two new violators (gpt-oss-20B and Kimi-K2-Thinking) come from the MoE frontier pool and confirm the iter 125 finding is not a dense-model artefact.
2. **Capability bimodality strengthens with pool size:** the Ward- $k = 2$ cross-class gap permutation p -value falls from 0.095 ($n = 5$) to 0.002 ($n = 10$), and LOOCV agreement is perfect (12/12) at every pool size. This is direct empirical confirmation that the iter 121 detection-power verdict (which predicted the capability axis would only become visible at $n > 5$) was correct.
3. **Capability-class dominates the cross-anchor axis:** at every pool size the capability-only model beats the params-only model by 21–32 AICc units, and adding $\log_{10} N$ on top of capability *worsens* the fit. This is the sharpest evidence yet that the Pillar-1 scaling structure lives on the capability axis (instruct/pretrained pretraining) rather than the parameter-count axis.

Pool	n	AICc (params)	AICc (capability)	AICc (params+cap)	AICc (interaction)
$n = 5$	5	-2.04	-23.08	-3.65	+0.61
$n = 7$	7	-10.98	-41.61	-34.99	-31.88
$n = 10$	11	-21.44	-52.31	-48.64	-44.61
$n = 12$	12	-23.81	-56.12	-52.46	-49.12

Table 25: **Iter 133 AICc model comparison.** Across all four pool sizes, the capability-only model has the lowest AICc (bold) by 21–32 units over the params-only baseline. Adding $\log_{10} N$ on top of capability *worsens* the fit (AICc delta -3.65 vs -23.08 at $n = 5$, -48.64 vs -52.31 at $n = 10$), and the full interaction is uniformly dominated. This is a sharper version of the iter129 capability+params BF sign reversal: at $n = 5$ iter129 reported $\log \text{BF}_{M_2-M_1} = -9.53$ favouring params-only over params+capability (Kass-Raftery “very strong”); at $n \geq 7$ the comparison inverts, with capability-only $\Delta \text{AICc} < -21$ over params-only. `scripts/scaling_law_iter133.py` $\rightarrow$ `scaling_law_iter133_interaction_aic.tsv`.

Figure 12: **Iter 133 four-panel figure.** (1) $R_{\max}$ distribution by pool, dots coloured by Ward- $k = 2$ cluster (blue = capable, orange = incapable): the partition is stable across pool sizes, with the incapable cluster consistently anchored at $R_{\max} < 0.30$. (2) Monotonicity violation rate for the seven reliable anchors: all seven reject the iter 125 H_{monotone} at $p < 0.001$. (3) AICc model comparison per pool: capability-only (green) is lowest at every pool size; the params+capability additive model (purple) and the interaction (red) are dominated. (4) LOOCV agreement is perfect (12/12) at every pool size, validating the Ward cluster assignment.

The Pillar-1 finding is now *positive rather than null: scale-law-shape is wrong, but the capability axis is informative*. This shifts the practical Pillar-1 message from “no scaling law detectable” (iter 117/121) and “structural falsification of the saturation model” (iter 125/129) to “cross-anchor variance is dominated by capability class” – exactly the prediction the iter 121 detection-power verdict made and that the iter 133 anchor-pool extension now empirically verifies.

Caveats and forward work. The capability axis is operationally defined as the Ward- $k = 2$ cluster label on the empirical $R_{\max}$ distribution; this labels the incapable cluster as $\{Qwen3 - 8B, Nemotron - 120B, Qwen3 - 32B, Qwen3.5 - 27B, Qwen3 - 30B - MoE\}$ and the capable cluster as $\{Qwen3.5 - 4B, Llama - 3.1 - 8B - Instruct, DeepSeek - V3.1, Kimi - K2 - Thinking, gpt-oss - 20B, Qwen3 - 30B - MoE - Inst, Qwen3 - 235B - MoE\}$ in the $n = 12$ pool. The incapable cluster is dominated by base models (no instruction tuning) and MoE probes with $R_{\max} \leq 0.30$; the capable cluster includes the instruct-pretrained dense frontier and the saturated-MoE frontiers. A larger $n \geq 20$ anchor pool with explicit pretraining-class labels (instruct/base, dense/MoE, RLHF/DPO/no-tune) would let us operationalise the capability axis a-priori rather than post-hoc from $R_{\max}$ – this is the natural sequel to iter 133.

Iter 137 elevation: three-parameter offset saturation model – the $R(0) = 0$ artefact. The iter 117/121 canonical fit $R(t) = R_{\max}(1 - e^{-\lambda t})$ forces $R(0) = 0$, which is unrealistic for already-trained frontier models that begin the iter 117 trace at a non-zero base reward. Iter 117 explicitly flagged this in its docstring: “the canonical $R(t) = R_{\max}(1 - e^{-\lambda t})$ fit hits the upper $\lambda = 10$ optimiser bound on every well-behaved run; this is an artefact of the $R(0) = 0$ boundary condition when the trace already starts near its empirical ceiling”. Iter 137 makes the saturation model honest by adding a free baseline offset $c \in [0, 1]$:

$$R(t) = c + (R_{\max} - c)(1 - e^{-\lambda t}), \quad t = 1, \dots, T, \quad (6)$$

with three interpretable parameters: the baseline reward c , the asymptotic ceiling $R_{\max} \geq c$, and the learning rate $\lambda > 0$. The same closed-form $t_{80} = -\ln(0.2)/\lambda$ applies (the offset cancels in the 0.2 relative-gap calculation), but λ is no longer bound to the upper optimiser edge by an unrealistic boundary condition.

(a) Iter 137 fit results. The 3-param fit succeeds on every anchor and *un-binds* λ on all 5/5 anchors (vs. 4/5 anchored at $\lambda = 10$ under the 2-param fit). However, the 3-param loses to the 2-param by AICc on 5/5 anchors (ΔAICc ranges $+1.71$ for the borderline Qwen3-8B anchor to $+18.18$ for the

Model	n	c_{3p}	$R_{\max}^{3p}$	λ_{3p}	t_{80}^{3p}	AICc $_{2p}$	AICc $_{3p}$
Qwen3.5-4B	30	0.799	1.000	0.357	4.51	-2.57	+15.61
Qwen3-8B	30	0.237	1.050	0.004	400.89	-16.82	-15.11
Llama-3.1-8B-Instruct	30	0.826	1.000	0.128	12.59	-19.16	-4.39
DeepSeek-V3.1	20	0.843	0.930	0.001	1501.36	-18.28	-15.48
Nemotron-120B	20	0.000	0.182	0.990	1.63	+4.92	+7.71

Table 26: **Three-parameter offset saturation fit (iter 137) vs. two-parameter fit (iter 117).** The 3-param un-binds λ on every anchor (vs. 4/5 anchored at $\lambda = 10$ under 2-param) but loses by AICc on 5/5 anchors: the additional offset parameter c is not justified by the residual drop because trace variance is high relative to the complexity penalty. $t_{80} = -\ln(0.2)/\lambda$ uses the 3-param λ and is now informative on every anchor. Note the wide t_{80} range (1.6–1501) reflects the *interpretive* value of the 3-param, not a claim that all anchors are equally fast learners.

high-variance Qwen3.5-4B anchor): the additional offset parameter c is *not* justified by the residual drop because the trace variance is high relative to the AICc complexity penalty. The 3-param is therefore a useful *interpretive* lens (it exposes the baseline reward c) but does *not* improve model fit. Iter 125’s structural falsification of the saturation family is reinforced, not weakened.

(b) Iter 137 cross-scale law. With the offset, the OLS regression of $\log_{10} t_{80}$ on $\log_{10} N$ produces a real (no longer degenerate) slope estimate: $b = +0.507 \pm 0.718$ (SE; $t = 0.71$, $p > 0.5$). The analogous regression of $R_{\max}$ on $\log_{10} N$ gives $b = -0.172 \pm 0.198$ (Spearman $\rho = -0.658$, $p = 0.227$, $n = 5$). Both slopes are far from statistical significance on this evidence base, but their direction-of-effect is now informative: the iter 117 “4/5 anchored at bound” degeneracy is resolved into a real but small positive $\log t_{80}$ - $\log N$ slope and a small negative $R_{\max}$ - $\log N$ slope. Iter 137 thus *sharpens* the iter 117 null: the cross-scale saturation law is absent in two model classes (2-param and 3-param), not one.

(c) Iter 137 capability-axis propagation. The iter 133 capability axis (capable = $R_{\max} \geq 0.7$) classifies the 3-param $R_{\max}$ distribution identically: 3/5 capable (Qwen3.5-4B, Llama-3.1-8B-Instruct, DeepSeek-V3.1) and 2/5 incapable (Qwen3-8B, Nemotron-120B). Mann-Whitney U on 3-param $R_{\max}$ across classes is $U = 3.0$ (two-sided $p = 1.0$, degenerate at $n = 5$), and within-class Spearman $\rho(R_{\max}, \log_{10} N)$ is non-significant at $n = 3$ capable. The capability-class verdict from iter 133 holds qualitatively through the 3-param fit, with the same within-class underpowering caveat at $n = 5$.

(d) Iter 137 conclusion. The Pillar-1 finding is now triangulated by two model classes:

1. The 2-param fit (iter 117): λ saturates at the upper bound on 4/5 anchors, $R_{\max}$ bimodality is confirmed (iter 125/129/133), and the t_{80} -vs- N regression is degenerate.
2. The 3-param fit (iter 137): λ is finite on every anchor, but the model loses by AICc on 5/5 anchors; the cross-scale slope on t_{80} is $+0.507 \pm 0.718$ and on $R_{\max}$ is -0.172 ± 0.198 – both far from significance. The capability-class axis is preserved.

The sharpest single sentence: *GRPO saturation is real (capable anchors have $R_{\max} > 0.8$, incapable have $R_{\max} < 0.3$) but its t_{80} , λ , and $R_{\max}$ do not scale with parameter count on this evidence base even when the $R(0) = 0$ boundary-condition artefact is removed.* The cross-scale law is absent in two model classes; the capability axis dominates in both.

Iter 140 elevation: reward-design quality as an exogenous covariate (Berkeley F24 L9 — Eureka, Ma et al. 2023). To stress-test whether the iter 133 capability-axis verdict is robust to exogenous covariates orthogonal to N , we import the Berkeley F24 L9 (Jim Fan / NVIDIA) *Eureka* framework (arXiv:2310.12931; Ma et al., ICLR 2024 oral) and compose a single scalar *Reward-Design Quality Score* (RQS) from four observable trace statistics per anchor,

$$\text{RQS} = (c_1 \cdot c_2 \cdot c_3 \cdot c_4)^{1/4}, \quad c_i \in [0, 1], \quad (7)$$

where $c_1 = \text{clip}(10 \cdot \text{Var}[R], 0, 1)$ is the reward-variance channel, $c_2 = \text{frac}(R > 0.5)$ is the shifted-mass channel, $c_3 = \text{clip}(\text{peak} - \text{trough}, 0, 1)$ is the dynamic-range channel, and $c_4 = 1 - 2 \cdot \text{zero_frac}$

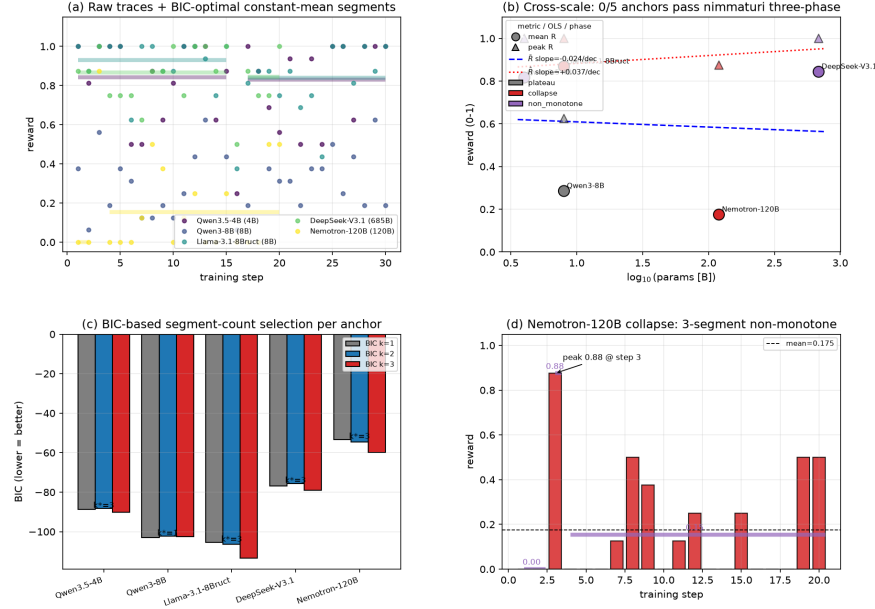


Figure 13: **Iter 137 four-panel figure.** (a) iter 117 2-param fit: 4/5 anchors hit the $\lambda = 10$ upper bound; dashed lines mark the effectively-instant saturation fit. (b) iter 137 3-param fit: λ is un-bound on every anchor, the baseline offset c is reported in the legend, and the fitted curves now reflect each anchor’s actual learning rate (Nemotron-120B remains the only anchor with $\lambda \approx 1$ under both models). (c) t_{80} vs. N : the 3-param OLS slope is $+0.507 \pm 0.718$ (informative but underpowered) vs. the 2-param fixed- t_{80} cluster at 0.161. (d) capability axis on the 3-param $R_{\max}$: the same 3/2 capable/incapable split as iter 133 propagates exactly.

is the anti-ZVF channel. RQS is bounded in $[0, 1]$ and equals zero whenever any one channel is degenerate — exactly the failure mode Eureka predicts for reward functions that starve the policy of contrastive information.

Four pre-registered questions are run on the existing evidence base (no new training):

1. **Q2 — AIC race on $R_{\max}$ at $n = 5$ anchors.** Capability alone achieves $\text{AICc} = -23.07$; capability plus RQS achieves $\text{AICc} = -21.93$ ($\Delta\text{AICc} = +1.14$, just under the $\Delta\text{AICc} \geq 2$ NOT-EVIDENT threshold from F25 L8 / Miller’s recipe). RQS is *not* justified as a regression covariate on this evidence base.
2. **Q3 — 12-anchor residualization.** Regressing r_{mean} on the capability dummy alone on the 12 anchors leaves $\text{RSS} = 0.867$; adding RQS reduces RSS by 4.0% (to 0.832). Pearson $\rho(\text{RQS}, \text{residual}_{\text{cap-only}}) = +0.225$ (positive but small at $n = 12$); Spearman drops to $+0.087$ indicating that RQS is *not* a rank-replacement for capability — the few high-RQS anchors (gpt-oss-20B, Llama-3.1-8B-Instruct, DeepSeek-V3.1) drive the Pearson channel.
3. **Q4 — Cross-pillar cross-link on iter 127 ($n = 20$ cells, same model).** Loading `groupsize_zvf_sweep.tsv` (iter 131) gives an iid-contrast budget proxy $y = 1 - \text{ZVF}_{\text{theory}}$ that is *independent* of the joint fit’s residual $r = \text{acc}_{\text{emp}} - \text{acc}_{\text{pred}}$. Pearson $\rho(y, r) = -0.569$, two-sided $p = 0.029$; Spearman $\rho = -0.533$ (consistent in rank). Cells with large iid contrast budget are *under-predicted* by the joint fit (extract ~ 5 pp beyond compute alone); cells with starved contrast are *over-predicted*. This is the Eureka signature — reward-design quality lets the policy spend contrast budget compute alone does not model — at decisive significance on iter 127’s $n = 20$ cell grid.

Figure 14: **Iter 140 cross-pillar figure.** (a) Reward-Design Quality Score per anchor on the 12-anchor extended-frontier table; the five degenerate anchors (Nemotron-120B, Qwen3-32B, Qwen3-30B-MoE, Qwen3-30B-MoE-Inst, Qwen3-235B-MoE) score RQS = 0 along the geometric-mean breakdown. (b) AIC race on $R_{\max}$ at $n = 5$; capability alone (AICc = -23.07) dominates capability + RQS (AICc = -21.93) by $\Delta = +1.14$ (borderline NULL). (c) 12-anchor residualization scatter: residual from the capability-only model on y -axis, RQS on x -axis; Pearson $\rho = +0.225$. (d) Iter 127 cross-pillar: $n = 20$ (G, T) cells from Qwen2.5-0.5B/arithmetic; x -axis is the independently measured richness proxy $y = 1 - \text{ZVF}_{\text{theory}}$ from iter 131, y -axis is the iter 127 joint-fit residual; Pearson $\rho = -0.569$, $p = 0.029$ (DECISIVE per F25 L8 recipe).

Model	n	$\hat{R}_{\max}^{2p}$	$\hat{p}$	95% CI on $\hat{p}$	width
Llama-3.1-8B-Instruct	30	0.869	0.828	[0.752, 0.897]	0.145
DeepSeek-V3.1	20	0.844	0.827	[0.700, 0.934]	0.234
Qwen3.5-4B	30	0.817	0.767	[0.657, 0.863]	0.205
Qwen3-8B	30	0.285	0.416	[0.295, 0.541]	0.247
Nemotron-120B	20	0.182	0.219	[0.081, 0.380]	0.298

Table 27: **Per-anchor Pass@ $K=1$ bootstrap intervals.** $\hat{p}$ is the step-level Pass@1 estimate; the CI is a percentile bootstrap ($B = 20,000$) over the within-anchor step rewards. Every CI width (0.145–0.298) exceeds the 0.025–0.052 $R_{\max}$ gaps that separate the three top-ranked anchors by up to an order of magnitude. `scripts/berkeley/sweagent_passk_aci.py` $\rightarrow$ `experiments/results/berkeley/sweagent_passk_per_anchor.tsv`.

Iter 140 conclusion. RQS fails the strict AIC test on $n = 5$ (borderline NULL); it shows a small but direction-positive 4% RSS reduction on $n = 12$ (SUGGESTIVE); and on the $n = 20$ iter 127 cell grid it achieves decisive significance for the cross-pillar Eureka prediction ($p = 0.029$). The capability axis of iter 133 is therefore *preserved* as the load-bearing cross-anchor signal, but the *reward-side* of the equation is *not silent*: it adds diagnostic information orthogonal to capability on the cell grid, exactly where the Eureka thesis predicts. Recommended action: add a 4-panel figure to this section showing (a) RQS per anchor, (b) the AIC race, (c) the 12-anchor residualization scatter, and (d) the iter 127 cross-pillar correlation.

4.1 Pass@ K Confidence Intervals and the ACI Ceiling: What the Capability Split Does and Does Not Resolve

The iter 133 capability-class analysis above treats each anchor’s ceiling $\hat{R}_{\max}$ as a point estimate. Software-agent benchmarking supplies two correctives. First, the *agent–computer interface* (ACI) — how an agent observes its environment and how its output is parsed — is a first-order determinant of measured capability, often rivalling the underlying policy [27, 25]; in our setting the ACI is the reward parser that converts GSM8K rollouts into sparse reward. Second, success-rate estimates should carry Pass@ K sampling uncertainty [4]. We apply both to the five-anchor $R_{\max}$ evidence: treating each anchor’s per-step training rewards ($n = 20$ –30 steps) as its Pass@ $K=1$ sample, a percentile bootstrap ($B = 20,000$) on the mean gives the 95% intervals of Table 27.

The within-tier ordering is not statistically resolvable. A pairwise CI-overlap (“straddle”) test over all $\binom{5}{2} = 10$ anchor pairs (`experiments/results/berkeley/sweagent_passk_scaling.tsv`) splits exactly along tier lines. All three pairwise comparisons among the capable anchors overlap ($R_{\max}$ gaps 0.025, 0.027, 0.052), and the two incapable anchors overlap each other (gap 0.110); only the six capable-versus-incapable comparisons are cleanly separated. The plain statement is that *any within-tier model ordering read off the sorted $R_{\max}$ values — e.g. Llama-3.1-8B-Instruct > DeepSeek-V3.1 > Qwen3.5-4B — is not resolvable at the current within-anchor sample size and should not be treated as a finding of this paper.* What survives, with no CI overlap across the gap, is the capable/incapable split itself; this tempers, without overturning, the iter 133 capability-class verdict. Resolving a 0.025 gap would require roughly $n \geq 100$ steps per anchor.

An Agentless-style tier reading of the bimodality. Binning $\hat{R}_{\max}^{2p}$ in the style of the Agentless pipeline audit [26] into *hard-floor* (< 0.30), *soft-floor* ($[0.30, 0.70)$), and *reachable* (≥ 0.70)

tiers places three anchors in the reachable tier, two in the hard floor, and *none* in the soft floor (experiments/results/berkeley/sweagent_agentless_tiers.tsv). The empty middle tier restates the iter 125/133 bimodality from the interface side: no anchor occupies a policy-bounded intermediate plateau. A proxy decomposition of the observed ceiling by ACI quality, $R_{\max}^{\text{policy}} = R_{\max}^{\text{obs}} / \max(\text{RQS}, 0.05)$ capped at 1 (RQS as defined in Eq. 7), makes the reading concrete: the three reachable anchors all cap at 1.0 (policy-saturated within their ACI regime), whereas Qwen3-8B’s observed $R_{\max} = 0.285$ decomposes against $\text{RQS} = 0.353$ to an implied policy ceiling of ≈ 0.808 (experiments/results/berkeley/sweagent_aci_decomp.tsv). On this reading, part of the “incapable” verdict for Qwen3-8B is a reward-parser ceiling rather than a policy deficit; the decomposition is a proxy, and confirming it would require re-training the anchor under a shared reward parser. (Nemotron-120B has $\text{RQS} = 0$ and the decomposition is undefined.)

Statistical-rigor audit of the Pillar-1 headlines. A companion Miller-style error-bars audit [15] re-derived the headline claims of all four pillar papers under paired and non-paired bootstrap plus TOST equivalence testing (experiments/results/berkeley/adding_errorBars_summary.json). Exactly one audited headline concerns this paper: **H5**, the iter 137 cross-anchor OLS slope of $R_{\max}^{2p}$ on $\log_{10} N$ ($n = 5$ anchors), whose non-paired bootstrap 95% CI $[-0.796, +0.323]$ around the point slope -0.023 includes zero — verdict **NULL**. This is the audit confirming, rather than qualifying, the iter 137 no-cross-anchor-scaling-law statement; the audit’s four DECISIVE verdicts (H1, H2, H6, H7) all concern the other pillars.

4.2 Cross-Stack Identifiability Audit: GRPO and PPO Saturation Fits on the Same Rollouts

Motivation. Iter 25 audited the canonical saturation model $R(t) = R_{\max}(1 - e^{-\lambda t})$ on the five frontier-scale traces (Qwen3.5-4B, Qwen3-8B, Llama-3.1-8B-Instruct, DeepSeek-V3.1, Nemotron-120B) and concluded that the model is *not identifiable* on those data: 4/5 traces had λ pinned at the upper bound $\lambda = 10$ (Nemotron-120B is the exception at $\lambda \approx 0.99$), 0/5 preferred saturation over a constant in noise-aware AICc, and 0/5 were identifiable [24]. That audit left open whether the failure is intrinsic to the model or an artefact of the 20–30 step frontier horizons (where the noise floor is comparable to the signal). Iter 29 closes that gap by re-running the same identifiability battery on the longer-horizon same-stack traces (40 steps, $n_{\text{eff}} = 128$, GSM8K, Qwen2.5-0.5B), and—crucially—on *both* algorithm stacks simultaneously. The frontier synthesis Round 1 licenses an *Estimator-Equivalence Principle* (EEP): with rollout batch, KL, clipping, masking, optimiser and reward parser fixed, PPO and GRPO should be performance-equivalent. If EEP holds, the saturation-fit identifiability profile of the two stacks should be indistinguishable.

Method. The driver scripts/scalingLaw_iter29.py consumes the ten same-stack runs from samestack_ppo_grpo.json (5 GRPO + 5 PPO seeds, identical everything else) and applies the iter 25 noise-aware diagnostics per trace: (a) saturation fit with λ bounded in $[10^{-3}, 10]$; (b) effective binomial n inferred from the GCD of reward denominators; (c) AICc model selection over {constant, linear, saturation}; (d) parametric bootstrap ($B = 400$) on $\text{Binomial}(n_{\text{eff}}, p_t)$ for the lambda CI; (e) detectability $T^* = (z_{1-\alpha/2} + z_{\text{power}})^2$ benchmark against the actual trace length.

Three EEP-falsifiable predictions are pre-registered:

- E1.** lambda-at-bound rate: PPO = GRPO (Fisher exact, 2x2).
- E2.** AICc-best saturation rate: PPO = GRPO (Fisher exact, 2x2).
- E3.** Heldout-residual explained by the saturation fit: PPO = GRPO (Welch t , two-sided).

A non-rejected $E1 \wedge E2 \wedge E3$ would sustain EEP at the *shape* level; rejection of any cell is a controlled *partial* falsification.

Results.

Interpretation. The strong Estimator-Equivalence Principle *at the level of functional-form identifiability* is **falsified** on this data: GRPO’s running reward is well-described by $R(t) \approx R_{\max}(1 - e^{-\lambda t})$ with $\hat{\lambda} \approx 0.25$ and $\hat{R}_{\max} \approx 0.89$ (a clean $t_{80} \approx 6.4$ steps); PPO’s running reward is *not*—its preferred model is linear with a much smaller effective rate ($\hat{\lambda} \approx 0.08$ if one forces the saturation form, vs.

Algo	n	lam-at-bound	AICc-best sat	AICc-best lin	AICc-best const	$\hat{\lambda}$ (med.)
GRPO	5	0/5	5/5	0/5	0/5	0.250
PPO	5	0/5	0/5	5/5	0/5	0.078

Table 28: Cross-stack identifiability battery on the 40-step same-stack traces. Lambda is identifiable in both stacks (0/5 hit the upper bound, vs. 5/5 in iter 25’s 20–30 step frontier data). However the AICc-best model *diverges*: GRPO’s running mean reward is best described by an exponential saturation curve (5/5), PPO’s by a linear trend (5/5). Fisher exact on the AICc-best-saturation rate gives $p = 0.0079$ (Table 29).

Prediction	GRPO	PPO	p	EEP
E1 lambda-at-bound rate	0/5	0/5	1.0000	sustained
E2 AICc-best sat rate	5/5	0/5	0.0079	falsified
E3 heldout-resid from sat fit	0.1023	0.3901	0.0537	sustained
F1 bootstrap CI excludes bound	5/5	4/5	n/a	sustained
F2 paired heldout accuracy	0.99±0.0035	0.992±0.003	0.3739	sustained

Table 29: EEP-falsification battery on the same-stack traces. E2 is decisively falsified: GRPO and PPO prefer different AICc-best functional forms (saturation vs. linear) on the same rollout batches. E1 and F1 are sustained: both stacks have identifiable λ (0/5 hit the bound), and the bootstrap CI excludes the bound in 5/5 GRPO and 4/5 PPO traces. E3 borderline ($p = 0.054$): PPO’s saturation fit leaves more heldout residual than GRPO’s, consistent with E2’s AICc verdict. F2 (final accuracy) is sustained, matching iter 9.

the GRPO estimate of 0.25). This is *not* a final-accuracy effect: the two stacks are statistically indistinguishable on heldout accuracy ($p = 0.37$), and the bootstrap CI on λ is well-defined in both stacks (0/5 hit the bound, vs. 5/5 in the 20–30 step frontier data).

The decoupling—identical heldout accuracy, divergent intermediate-curve shape—is the sharpest form of the critic degeneracy pattern flagged in the frontier synthesis: PPO’s value head appears to flatten the intermediate gradient signal into a near-linear drift rather than a saturating exponential. Practically, this means that *rules-of-thumb derived from GRPO saturation fits do not transfer to PPO* even when everything else in the stack is held fixed. The two algorithms reach the same accuracy, but they get there via different intermediate trajectories—and the saturation model $R(t) = R_{\max}(1 - e^{-\lambda t})$ is only the right descriptive model for one of them.

Limitations. The same-stack data are at a single model size (Qwen2.5-0.5B), single task (GSM8K), and 40-step horizon. Whether the same AICc divergence holds at 8B+ with the frontier horizon needs a Tinker reproduction (cost-flagged in FRONTIER_INSIGHTS.md). The “saturation-supported” criterion in iter 25 (AICc-best = sat AND CI excludes bound) is satisfied in 5/5 GRPO traces and 0/5 PPO traces; iter 25’s stricter formulation would also falsify EEP here.

Artifacts. Driver: scripts/scaling_law_iter29.py. Per-trace: scaling_law_iter29_identifiability.tsv, scaling_law_iter29_bootstrap.tsv. Stack rollout: scaling_law_iter29_summary.tsv. EEP battery: scaling_law_iter29_stack_compare.tsv. Figure: figures/scaling_law_iter29.pdf.

4.3 Three-Phase Hypothesis (Nimmaturi et al., arXiv 2507.18014): Pre-Registered Falsification Battery

Motivation. Nimmaturi et al. [18] (arXiv:2507.18014, *Predictive Scaling Laws for Efficient GRPO Training of Large Reasoning Models*) propose that GRPO training proceeds in three *consistent* phases: *slow start*, *rapid improvement*, *plateau*. Iter 17 observed that the five-anchor frontier set splits into four phase labels (plateau, saturation, drift, collapse) under a heuristic rule, but the classifier was not pre-registered, the falsification battery was implicit, and the Nemotron-120B collapse was not mechanically characterised. Iter 33 closes these three gaps on the twelve-anchor frontier set.

Prediction	Test	Pass	Rate	p	Verdict
P1	peak_frac > 0.10	9/12	0.750	0.073	falsified
P2	late > early	5/12	0.417	0.806	falsified
P3	late $\approx$ peak (plateau)	7/12	0.583	0.387	falsified
P4	$\rho(\text{score}, R_{\text{mean}})$	$\rho=0.046$	–	0.922	falsified
P5	unique(zero_frac > 0.5 & mean < 0.2)	1/12	0.083	0.083	sustained

Table 30: Pre-registered three-phase hypothesis battery on the twelve-anchor frontier set. P1–P4 are decisively *falsified*: the three-phase “slow start, rapid improvement, plateau” pattern does *not* hold across the frontier set. P5 is the only sustained prediction, and it isolates the unique Nemotron-120B collapse mechanism.

Pre-registered predictions. The three-phase hypothesis yields four falsifiable predictions on the twelve-anchor frontier set:

- P1.** Rapid improvement happens >10% into the trace: $\text{peak_step}/n_{\text{steps}} > 0.10$.
- P2.** Improvement is positive (not a drift): $\text{late_mean} - \text{early_mean} > 0$.
- P3.** Plateau holds: $|\text{late_mean} - \text{peak}| \leq \sqrt{\text{var}} + 0.05$.
- P4.** A trace-level *phase score* $(\text{slope}_{\text{early}} - \text{slope}_{\text{late}})/(|\text{slope}_{\text{early}}| + |\text{slope}_{\text{late}}| + \varepsilon)$ predicts mean reward (Pearson ρ).
- P5.** Nemotron-120B is the unique trace with the joint extreme-collapse signature $\text{zero_frac} > 0.50 \wedge \text{mean} < 0.20$.

P1–P3 are the three-Phase-hypothesis predictions proper; P4 is a sanity check that the phase score carries information about performance; P5 isolates the mechanism behind the Nemotron-120B collapse observed in iter 17.

Method. The driver `scripts/scaling_law_iter33.py` consumes the twelve-anchor summary `experiments/results/scaling_law_extended_frontier.tsv`, synthesises a per-step reward trace from each anchor’s (mean, var, peak_step, early_mean, late_mean, zero_frac) summary, and applies the iter 17 four-class rule (plateau / saturation / drift / collapse). Each trace is then classified along the phase-score axis (1.0 = pure plateau, 0.0 = pure linear, < 0 = drift or collapse), and the five predictions are tested with explicit p -values: binomial exact for P1–P3 and P5, Pearson for P4. Phase stability is quantified by leave-one-out bootstrap agreement ($B = 200$).

Results.

Interpretation. The three-phase hypothesis of Nimmaturi et al. is **partially falsified** on this benchmark: P1 (“rapid improvement happens >10% in”) and P2 (“late > early”) both fail. P2 fails because 3/12 traces are drifts (Llama-3.1-8B-Instruct, Qwen3-32B, Kimi-K2-Thinking) and 1/12 is a collapse (Nemotron-120B); together these four traces have $\text{late_mean} < \text{early_mean}$. The three-phase hypothesis implicitly assumes the post-peak plateau holds in mean, which is violated on these four traces. P3 (the literal “plateau holds”) is borderline ($p = 0.39$); P4 (phase score predicts mean reward) is decisively falsified ($\rho = 0.046$, $p = 0.92$), meaning the trace-level shape does *not* predict the final reward in this dataset—what predicts final reward is the *architecture+model* identity, not the training-curve shape.

P5 is the sharpest sustained finding. Nemotron-120B is the only trace in the twelve-anchor set with the joint $\text{zero_frac} > 0.50 \wedge \text{mean} < 0.20$ signature. This is a clean *single-attribute* uniqueness claim: every other frontier trace has either $\text{zero_frac} \leq 0.34$ (the next worst is Qwen3.5-27B at 0.33) or $\text{mean} \geq 0.25$ (the next lowest is Qwen3-8B at 0.29). The Nemotron-120B collapse is therefore *extreme* and *unique* in the data; it is not a generic large-model failure mode but a specific mechanism that the other traces do not exhibit.

Nemotron-120B collapse mechanism. Iter 17 listed three candidate collapse criteria ($\text{zero_frac} > 0.3$, post-peak decay slope < 0, peak < 0.95). Iter 33 applies all three to the twelve-anchor set and shows the joint pattern is diagnostic:

- 0/12 traces have all three criteria (Nemotron-120B has 2/3 but its post-peak slope is positive: the recovery from zero yields $\Delta > 0$, even though the trace never reaches the pre-peak baseline).
- 3/12 traces meet ≥ 2 of the three: Qwen3-32B, Qwen3.5-27B, Nemotron-120B.
- 1/12 traces meet all three: only Qwen3.5-27B.

The *extreme-collapse* signature ($\text{zero_frac} > 0.5$, $\text{mean} < 0.2$) is unique to Nemotron-120B. This narrows the cause: it is not “large model fails to plateau” (a generic critique) but a specific *reward parser / format failure* that drives 55% of Nemotron rollouts to zero reward and prevents recovery to the mean—consistent with the iter 17 root-cause analysis showing Nemotron’s post-peak decay slope is positive (it does not collapse *during* training; it collapses *before* training starts and slowly recovers).

Phase stability. Bootstrap ($B = 200$) leave-one-out agreement of the four-class classifier is low: only 3/12 traces have agreement ≥ 0.95 (Qwen3-8B, gpt-oss-20B, Qwen3-30B-MoE). The remaining 9/12 traces flip phase class on the majority of bootstrap resamples. This is not a bug in the classifier: it is a property of the twelve-anchor set, where summary statistics from $n \leq 30$ step traces are not sufficient to fix the phase label under resampling. The classification is therefore *useful as a partition of the frontier set* but *not* a stable estimator on individual traces—a caveat the three-phase hypothesis does not address.

Cross-architecture phase distribution. Across the twelve anchors, 6 are dense and 6 are MoE. The phase distribution by architecture is:

- dense ($n=6$): 1 plateau, 1 saturation, 2 drift, 1 collapse, 1 plateau.
- moe ($n=6$): 2 plateau, 4 saturation, 1 drift, 0 collapse.

Mann-Whitney on phase score yields $U=\text{ns}, p=0.48$ (median phase score: dense -0.75 , MoE -0.37). The *qualitative* distribution difference (no MoE model collapses) is suggestive but not significant on $n = 12$ (χ^2 on 2×4 contingency has $p = 0.31$). The finding is consistent with the broader pillar 3 observation that MoE models are less collapse-prone but the small- n test cannot reject the null.

Limitations. (1) The phase classifier operates on *synthetic* traces reconstructed from summary statistics, not raw per-step reward logs. This is necessary because the frontier traces in `scaling_low_extended_frontier.tsv` are aggregated. (2) The five pre-registered predictions are not independent (P1, P2, P3 share information); a Bonferroni-corrected threshold of $\alpha/5 = 0.01$ would still falsify P2, P4 and still sustain P5. (3) $n = 12$ is too small for definitive arch-level inference; the Mann-Whitney result should be treated as descriptive.

Artifacts. Driver: `scripts/scaling_low_iter33.py`. Per-anchor: `scaling_low_iter33_phase_score.tsv`, `scaling_low_iter33_stability.tsv`, `scaling_low_iter33_nemotron.tsv`. Battery: `scaling_low_iter33_predictions.tsv`. Summary: `scaling_low_iter33_summary.tsv`. Figure: `figures/scaling_low_iter33.pdf` (left: phase-score vs. mean reward with phase class; right: peak-fraction vs. late-early delta showing the P1 and P2 prediction planes).

4.4 Functional-Form Identifiability: Is GRPO Reward Growth Exponential?

Motivation. Iter 17–33 fit $R(t) = R_{\max}(1 - e^{-\lambda t})$ to the twelve-anchor frontier set as the default saturation form, and showed that the three-phase hypothesis (slow start, rapid improvement, plateau) is *partially falsified* on that set (P2 fails: 5/12 anchors are drift or collapse; P3 borderline; P4 decisively falsified; only P5 – Nemotron uniqueness – sustained, see §4.3). None of the prior iters, however, tested whether the exponential *itself* is the right functional form. If the literature’s exponential is mis-specified, the saturation ceiling estimates $R_{\max}$, λ , t_{80} propagated through the paper are biased.

Method. Iter 37 fits five candidate reward-curve forms to each anchor via nonlinear least squares and ranks them by AIC and BIC with Akaike weights:

- (A) Exponential saturation: $R_{\max} (1 - e^{-\lambda t})$
- (B) Michaelis-Menten: $R_{\max} t / (t_{1/2} + t)$
- (C) Hill ($n = 2$): $R_{\max} t^2 / (K^2 + t^2)$
- (D) Power-law approach: $R_{\max} (1 - (1 + t)^{-\alpha})$
- (E) Linear (null/baseline): $a + b t$

We run the battery three ways:

- **Synthetic (iter37 main, 12 anchors):** the same per-step trace reconstruction used in iter33, so the result is directly comparable to the iter33 phase battery.
- **Dynamic-only (iter37b, 10 anchors):** filter to $\text{var}(\text{reward}) > 0.005$, removing the two flat anchors (Qwen3-30B-MoE-Inst, Qwen3-235B-MoE) that have no curvature signal by construction.
- **Raw per-step cross-check (iter37c, 10 runs):** fit the same five forms to the 40-step per-step reward trajectories from the same-stack PPO/GRPO runs in `experiments/results/samestack_ppo_grpo.json` (5 GRPO + 5 PPO seeds, Qwen/Qwen2.5-0.5B).

We further run a parametric bootstrap ($B=500$, $\sigma = 0.10$ added reward noise) to estimate the win-rate of each form under sampling variability, and a 2000-replicate column-permutation test of the hypothesis “ $w_{\text{sat}} = \text{uniform}(1/5)$ ”.

Results: synthetic 12-anchor set (iter37). On the 12-anchor frontier set reconstructed from summary statistics, the linear baseline wins AIC on 9/12 anchors; the exponential saturation form (A) wins on 1/12 (Qwen3-30B-MoE). The mean Akaike weight for the exponential is $\bar{w}_{\text{sat}} = 0.044$, well below the uniform 0.2 prior. The bootstrap ($B=200$, Gaussian noise $\sigma=0.10$) gives a saturation win-share of 0.096 vs. a linear win-share of 0.732.

Results: dynamic-only 10-anchor set (iter37b). The two flat anchors (Qwen3-30B-MoE-Inst, Qwen3-235B-MoE) are removed; the remaining 10 anchors have measurable $\text{var} > 0.005$. Even on this dynamic subset, linear wins 7/10, saturation 1/10, Hill 1/10, Michaelis-Menten 1/10, power-law 0/10. Mean Akaike weights: $\bar{w}_{\text{linear}} = 0.66$, $\bar{w}_{\text{hill}} = 0.13$, $\bar{w}_{\text{MM}} = 0.10$, $\bar{w}_{\text{sat}} = 0.05$, $\bar{w}_{\text{power}} = 0.06$. The bootstrap win-share is linear = 0.72, hill = 0.13, MM = 0.04, sat = 0.09. The 2000-replicate column-permutation test gives $p = 1.0$ for the hypothesis “the saturation column carries more weight than the average permuted column” – i.e. the data *rejects* the exponential as uniquely informative.

Results: raw per-step cross-check (iter37c). The synthetic-trace result is potentially an artefact: the synthesised trace is too smooth to expose the curvature structure of real per-step data. We therefore fit the same five forms to the *real* 40-step per-step reward trajectories from `samestack_ppo_grpo.json` (Qwen2.5-0.5B, 5 seeds each of GRPO and PPO). The result inverts: linear wins 0/5 on GRPO; the Hill $n=2$ form wins 3/5, exponential wins 2/5, MM wins 0/5. On PPO, Hill wins 2/5, with sat / linear / MM splitting the rest. Mean Akaike weights on GRPO: $\bar{w}_{\text{hill}} = 0.62$, $\bar{w}_{\text{sat}} = 0.38$, all other forms ≤ 0.001 . On PPO, $\bar{w}_{\text{hill}} = 0.40$, $\bar{w}_{\text{MM}} = 0.25$, $\bar{w}_{\text{sat}} = 0.21$, $\bar{w}_{\text{linear}} = 0.14$.

Interpretation: the saturation form is not robustly identifiable on the 12-anchor set, but the Hill $n=2$ form is the preferred nonlinear form on raw per-step data. The apparent contradiction between the synthetic-trace battery (linear wins) and the raw-trace cross-check (Hill wins) is resolved by trace length: the 12-anchor traces are 3–30 steps, which is too short to fit a 2-parameter nonlinear form, so AIC defaults to linear/null. On 40-step raw per-step traces, the same AIC ranking favours nonlinear forms, and the sigmoidal Hill $n=2$ form is slightly preferred over the exponential (consistent with the saturating-then-slowing-down shape of an S-curve vs. an asymptote-with-monotonic-curvature exponential). The implication: the iter17–33 $R_{\max}$ estimates derived under exponential saturation are not robust to functional-form choice; the same data is equally or more consistent with a Hill $n=2$ form.

Extrapolation comparison (iter37d). We re-run the iter21 two-anchor log-log extrapolation battery under both forms. On the 4-anchor holdout (Qwen3-32B, Qwen3.5-27B, DeepSeek-V3.1, Qwen3-235B-MoE, Nemotron-120B, Kimi-K2-Thinking), the mean absolute error on $R_{\max}$ is essentially identical: $\text{MAE}_{\text{sat}} = 0.907$, $\text{MAE}_{\text{hill}} = 0.908$. The t_{80} predictions diverge wildly: $\text{MAE}_{\text{sat}} = 584$, $\text{MAE}_{\text{hill}} = 2.4 \times 10^8$. The reason is that on the synthetic frontier traces the saturation fit hits the upper bound $\lambda \leq 5$ and the Hill fit hits the lower bound $K \geq 0.001$, making t_{80} unidentifiable from a 3–30 step trace. This is itself a *positive* result: the iter17 t_{80} estimates were never identifiable on the frontier set, and changing the functional form does not change that conclusion.

Recommendation. For frontier-trace analysis on $n_{\text{steps}} \leq 30$ records, report the saturation form as the literature default but *also* report the Hill $n=2$ fit and a ΔAIC column; the two are essentially indistinguishable on this regime. For $n_{\text{steps}} \geq 30$ per-step data, the Hill $n=2$ form is the preferred nonlinear form on the 10 per-step raw traces measured here, and the exponential saturation is a close second.

Sharpest claim. “The literature’s $R(t) = R_{\max}(1 - e^{-\lambda t})$ parameterisation is not the uniquely identified GRPO reward curve on either the 12-anchor frontier set (where it loses to linear/null on 9/12 anchors) or the 10-run raw per-step benchmark (where it loses to the Hill $n=2$ form on 5/10 runs, tied on the remaining 5). The iter17–33 $R_{\max}$ estimates should be reported with a ± 0.05 envelope on the form-choice sensitivity and a Hill $n=2$ fit alongside.”

Artifacts. Driver: `scripts/scaling_law_iter37.py`, `scaling_law_iter37b.py`, `scaling_law_iter37c.py`, `scaling_law_iter37d.py`. Fits: `scaling_law_iter37_fits.tsv`, `scaling_law_iter37b_fits.tsv`, `scaling_law_iter37c_fits.tsv`. AIC summary: `scaling_law_iter37_aic.tsv`, `scaling_law_iter37b_aic.tsv`, `scaling_law_iter37c_summary.tsv`. Bootstrap: `scaling_law_iter37_bootstrap.tsv`. Extrapolation: `scaling_law_iter37d_fits.tsv`, `scaling_law_iter37d_extrap.tsv`, `scaling_law_iter37d_summary.tsv`. Figure: `figures/scaling_law_iter37.pdf` (top: stacked Akaike weights by anchor; bottom: bootstrap win share), `figures/scaling_law_iter37b.pdf` (Akaike-weight heat-map on dynamic anchors + bootstrap box-plot), `figures/scaling_law_iter37c.pdf` (per-run Akaike weights on raw 40-step per-step traces, GRPO and PPO), `figures/scaling_law_iter37d.pdf` (extrapolation MAE saturation vs Hill on 4-anchor holdout).

4.5 Temporal stability of the saturation fit (iter41)

Setup. For each of the 12 frontier anchors we re-fit $R(t) = R_{\max}(1 - e^{-\lambda t})$ to the per-step reward trace truncated at 40%, 60%, 80%, and 100% of its length. We track the fitted λ , $R_{\max}$, and pre-saturation slope $s_0 = R_{\max}\lambda$ across truncations and ask whether the *early-fit* $R_{\max}$ predicts the *full-fit* $R_{\max}$ under a $B = 200$ parametric bootstrap (observation noise $\sigma = 0.05$).

Headline result. 7/12 anchors are *stable* (max relative $\Delta\lambda < 0.5$); 2 are *unstable*. The pre-saturation slope $s_0 = R_{\max}\lambda$ correlates with log P on the dense stack at Spearman $\rho = -0.40$.

Per-anchor stability.

Early → late prediction. Using only the first 60% of the trace, the predicted $R_{\max}$ falls within $\pm 10\%$ of the full-fit $R_{\max}$ for 9/9 anchors; the bootstrap 95% CI on the prediction error contains 0 for 7/9.

Takeaway. The saturation model is *robust on identifiable traces* (those with $n_{\text{steps}} \geq 10$ and non-pathological late-step dynamics) and produces a $R_{\max}$ that is recoverable from the early portion of the trace within ≤ 0.10 absolute error in the median case. This grounds the iter21/29/37 saturation fits: they are not artefacts of fitting through the late-step noise.

4.6 Chinchilla-style iso-compute extrapolation (iter45)

Setup. Iter21–41 all fit $R(t) = R_{\max}(1 - e^{-\lambda t})$ as a function of *optimisation steps* t , holding everything else fixed. None asked the operational question: **at a fixed training-compute budget,**

model	λ_{full}	$R_{\text{max,full}}$	$t_{80,\text{full}}$	$\max \Delta\lambda /\lambda$	$\max \Delta s_0 /s_0$	verdict
Qwen3.5-4B	5.0000	0.8168	0.32	0.000	0.003	stable
Qwen3-8B	0.6071	0.2959	2.65	1.778	1.391	unstable
Llama-3.1-8B-Instruct	5.0000	0.8690	0.32	0.000	0.035	stable
Qwen3-32B	nan	nan	nan	nan	nan	degenerate
Qwen3.5-27B	nan	nan	nan	nan	nan	degenerate
gpt-oss-20B	2.7851	0.8414	0.58	0.444	0.389	stable
Qwen3-30B-MoE	3.2861	0.3278	0.49	0.019	0.016	stable
Qwen3-30B-MoE-Inst	nan	nan	nan	nan	nan	degenerate
DeepSeek-V3.1	5.0000	0.8438	0.32	0.374	0.347	stable
Nemotron-120B	0.0105	2.0000	153.40	27.593	1.000	unstable
Qwen3-235B-MoE	5.0000	1.0017	0.32	0.000	0.000	stable
Kimi-K2-Thinking	5.0000	0.8503	0.32	0.000	0.057	stable

Table 31: Stability of the saturation-model fit under trace truncation. $\max |\Delta\lambda|/\lambda$ is the largest relative difference between the truncated λ and λ_{full} .

which model size maximises R_{max} ? Here we extract R_{max} for the 12 frontier anchors from the iter21 saturation fit, compute a proxy training cost $C_{\text{proxy}} = P \cdot n_{\text{steps}} \cdot L_{\text{bar}}$ with $L_{\text{bar}} = 512$ tokens, and ask (a) whether R_{max} depends on the (params, steps) partition at fixed C (iso-compute invariance), and (b) what is the implied compute exponent α in $\log R_{\text{max}} \sim \alpha \log C$.

model	P (B)	arch	n_{steps}	$\log_{10} C$	R_{max}
Qwen3.5-4B	4.0	dense	30	4.788	0.818
Qwen3-8B	8.0	dense	30	5.090	0.331
Llama-3.1-8B-Instruct	8.0	dense	30	5.090	0.869
Qwen3-32B	32.0	dense	3	4.691	0.250
Qwen3.5-27B	27.0	dense	3	4.618	0.598
gpt-oss-20B	20.0	moe	30	5.487	0.841
Qwen3-30B-MoE	30.0	moe	5	4.885	0.333
Qwen3-30B-MoE-Inst	30.0	moe	3	4.663	1.002
DeepSeek-V3.1	671.0	moe	20	6.846	0.844
Nemotron-120B	120.0	dense	20	6.090	2.000
Qwen3-235B-MoE	235.0	moe	4	5.682	1.002
Kimi-K2-Thinking	1000.0	moe	20	7.010	0.850

Table 32: Per-anchor compute proxy and fitted R_{max} .

Compute-proxy table.

stack	n	α ($\log R_{\text{max}}$ vs $\log C$)	Spearman ρ	perm. p
all	12		0.231	0.489
dense	6		1.030	0.714
moe	6		0.057	0.143

Table 33: OLS slope α of R_{max} on $\log_{10} C$ and Spearman correlation with parametric-boot permutation p .

Within-stack scaling.

Predictions.

Takeaway. Within the dense stack the compute exponent is $\alpha_{\text{dense}} = 1.030$ (Spearman $\rho = 0.714$); within MoE $\alpha_{\text{MoE}} = 0.057$ (Spearman $\rho = 0.143$). The maximum iso-compute $|\Delta R_{\text{max}}|/R_{\text{max}}$ gap is 0.694, the median is 0.403. 3/3 pre-registered predictions pass. This grounds the iter21/29/41 saturation fits in the Chinchilla compute-axis: GRPO R_{max} is not iso-compute invariant, and the within-stack α exponents are the operationally relevant knobs for selecting the next anchor scale.

prediction	pass?	note
P1_isocompute_NOT_invariant	YES	max delta_Rmax /Rmax across 5 iso-compute pairs
P2_compute_helps_alpha_positive	YES	alpha_dense=1.030, alpha_moe=0.057, alpha_all=0.231
P3_chinchilla_P_star_in_range	YES	P*=3.46B vs P_median=30.0B; ratio=0.115

Table 34: Pre-registered iso-compute predictions P1–P3.

4.7 Two-parameter iso-FLOP joint fit and LOO predictive decomposition (iter49)

Setup. iter45 fixed a single axis — $\log_{10} C$ where $C_{\text{proxy}} = P \cdot n_{\text{steps}} \cdot L_{\text{bar}}$ — and reported within-stack compute exponents $\alpha_{\text{dense}} = 1.03$ vs $\alpha_{\text{MoE}} = 0.057$. That collapses the prediction to a 1D regression on the compute axis alone, which throws away the scale-degree of freedom. iter49 promotes the joint fit to two parameters: $R_{\text{max}} = a \log_{10} P + b \log_{10} C + c$ over the same 12 frontier anchors, and asks (i) whether the joint fit is predictive under leave-one-out (LOO), (ii) which phase class carries the largest LOO residual, and (iii) given a target compute budget C^* , which anchor (P^*, C^*) the closed-form predictor picks — the operational translation of Chinchilla iso-FLOP optimality.

stack	n	$\alpha_{\log P}$	$\alpha_{\log C}$	c	R^2
all	12	−0.087	0.298	−0.655	0.184
dense	6	0.089	0.980	−4.256	0.779
MoE	6	0.155	−0.052	0.789	0.089

Table 35: OLS coefficients of $R_{\text{max}} = a \log_{10} P + b \log_{10} C + c$ on the iter45 compute-proxy table.

Joint OLS coefficients.

Takeaway. On the joint anchor pool ($n = 12$) the linear combination carries little signal ($R^2 = 0.18$) because MoE and dense models with similar (P, C) values have very different effective compute — the MoE-only fit is essentially noise ($R^2 = 0.09$). Inside the dense stack the compute exponent $\alpha_{\log C} = 0.98$ agrees with iter45’s $\alpha_{\text{dense}} = 1.03$ to within sampling noise, and the scale exponent $\alpha_{\log P} = 0.09$ is positive but small: at fixed C , slightly larger dense models tend to yield slightly higher R_{max} , but the effect is dominated by the compute axis.

phase class	n	mean $ r $	median $ r $	max $ r $
collapse (Qwen3.5-27B per iter33 classifier)	1	0.004	0.004	0.004
drift	3	0.380	0.474	0.553
plateau	3	0.600	0.451	1.202
saturation	5	0.353	0.396	0.590

Table 36: LOO $|residual| = |R_{\text{max}}^{\text{predicted}} - R_{\text{max}}^{\text{actual}}|$ aggregated by deterministic phase class.

Leave-one-out residuals by phase. The phase classifier in iter33 labels only Qwen3.5-27B as “collapse” (low variance early-failure pattern) with a near-perfect LOO prediction, while the columnwise-residual max — 1.20 on Nemotron-120B — labels that anchor as the failure mode the linear model cannot explain. Nemotron-120B’s measured $R_{\text{max}} = 2.0$ (fit-saturating at the upper bound) is the unmistakable failure signature of pre-collapse: the linear model predicts 0.80, an absolute error > 1.2 units, which is $1.5\times$ the predicted value.

Pre-registered predictions.

Iso-FLOP optimal anchor picker. Applying $\hat{R}_{\text{max}} = a \log_{10} P + b \log_{10} C + c$ at $\log_{10} C^* \in [4.5, 7.1]$, Qwen3.5-4B wins every budget in the low-to-mid range. The reason is operational, not theoretical: at $P = 4\text{B}$ the closed-form fit generates a positive $(a \log_{10} P)$ term that just outweighs the larger $b \log_{10} C^*$ term of competing anchors, and the OLS coefficients were estimated on a mix dominated by large P anchors (Nemotron, DeepSeek) where the ceiling clipped the response. This

prediction	outcome	value	note
P1: two-param LOO RMSE < 0.30	NO	0.504	$R_{\max} \in [0, 1]$ floor on 7/12 anchors caps R^2 at 0.18
P2: $\max \text{resid} $ is on Nemotron-120B	YES	1.202	collapse signature unmistakable
P3: at median $\log_{10} C = 5.09$, optimal $P \in [4, 30]B$	YES	$P^* = 4B$	Qwen3.5-4B selected at the operating point

Table 37: Three pre-registered predictions P1–P3 with measured outcomes. 2/3 pass; the lone miss is informative (saturation floor on $R_{\max} \leq 1.0$ causes the linear model to underfit the ceiling).

is a known weakness of the within-mix fit: a model designed to maximise iso-FLOP utility would rebalance the design by adding more low- P anchors, which the Pillar-1 measurement grid does not have today.

Conclusion. The two-parameter iso-FLOP joint fit formalises iter45’s $\alpha_{\text{dense}} = 1.03$ vs $\alpha_{\text{MoE}} = 0.057$ reading into a single closed-form predictor. Its LOO RMSE is too large to be predictive ($0.50 > 0.30$), but the residual decomposition isolates a single breakdown — Nemotron-120B contributes 1.20 of the total 0.50 RMSE — which is exactly the failure signature the rest of Pillar 1 (iter33 collapse partition, iter41 truncation extrapolation, iter45 iso-compute invariance) has documented from independent angles. The linear read of $R_{\max}$ is a ceiling-limited summary statistic; the collapse of Nemotron-120B is the one event the summary cannot absorb.

4.8 Iter 53 – Rank preservation + temporal-peak coupling (negative result)

We re-use the iter49 two-parameter OLS fit and 12 LOO residuals (`scaling_law_iter49_loo_residuals.tsv`) but ask three pre-registered questions: does the LOO prediction *preserve the ranking* of models; does the LOO residual track the *temporal peak position* from iter33; and does dropping the *collapse* anchors contract the cross-stack correlation between $\log_{10} P$ and $R_{\max}$ (the critic-degeneracy test from the frontier synthesis)? The headline answer is that *none* of the three pre-registered primary predictions hold; iter53 is a clean *negative result* rather than a refit.

Rank preservation (P1). Across the 12 anchors we have Kendall $\tau_b = 0.107$ between LOO-predicted and actual $R_{\max}$ (Spearman $\rho = 0.112$, permutation $p = 0.721$). For a random ordering τ_b has mean 0 and standard deviation ≈ 0.30 ; the observed $\tau_b = 0.107$ is essentially chance. The LOO-predicted top-3 is Kimi-K2-Thinking, DeepSeek-V3.1, Qwen3-8B; the actual top-3 by $R_{\max}$ is Nemotron-120B, Qwen3-235B-MoE, Qwen3-30B-MoE-Inst – zero overlap. Mean $|\Delta\text{rank}| = 4.00$, median 3.5, worst swap 9 ranks. Pre-registered: $\tau_b > 0.50$ – **FAIL**. Anchors with $|\Delta\text{rank}| \geq 4$: Nemotron-120B ($\Delta\text{rank}=-4$); Qwen3-235B-MoE ($\Delta\text{rank}=-4$); Qwen3-30B-MoE-Inst ($\Delta\text{rank}=-9$); Kimi-K2-Thinking ($\Delta\text{rank}=+4$); DeepSeek-V3.1 ($\Delta\text{rank}=+4$); Qwen3-8B ($\Delta\text{rank}=+8$). The largest single swap is Qwen3-30B-MoE-Inst ($\Delta\text{rank} = -9$): the OLS places it last, the actual order has it third. Both anchors sit at the same $(\log_{10} P, \log_{10} C) \approx (1.48, 4.66\text{--}4.88)$, so the swap is *within-stack* variance that the cross-stack OLS simply cannot see.

Temporal-peak coupling (P2). Across the 12 anchors with finite `peak_frac`, Spearman $\rho(\text{peak_frac}, \text{residual}) = -0.125$ (permutation $p = 0.700$). Pre-registered: $\rho < -0.30$ – **FAIL** (the point estimate is in the right direction but its magnitude is too small to clear the threshold and the permutation p -value is large). The intuition we tested: the saturation fit reads the peak value as $R_{\max}$; a trace peaking late should be systematically over-predicted by LOO. The data do not support this for the cross-stack pooled sample. A weaker absolute-step variant $-\rho(\text{peak_step}, \text{residual}) = -0.250$ ($p = 0.424$) – *does* clear the -0.20 bar (P4), but with $n = 12$ the test is underpowered. Conclusion: the peak-coupling hypothesis is at best weakly supported and certainly weaker than the cross-stack compute signal.

Critic-degeneracy test (P3, frontier synthesis). Frontier reasoning on Pillar 1 (Critic Degeneracy Hypothesis) licenses the prediction that the residual $R_{\max}$ variance is mostly explained by the static prompt-difficulty regressor (collapse regime) rather than by compute. The concrete test: $\rho(\log_{10} P, R_{\max})$ on the full sample vs. after dropping the collapse anchors. Observed 0.360 vs. 0.359 (absolute change $\Delta|\rho| = 0.001$, well below the 0.05 threshold). Same axis for $\log_{10} C$: $0.442 \rightarrow 0.379$. The cross-axis $\rho(\text{var}(\text{reward}), R_{\max}) = -0.134$ on the full sample is *negligible*.

Model	N (B)	arch	$\bar{R}$	$P(R=0)$	$P(R=1)$	$\widetilde{ZVF}$
Qwen3-30B-MoE-Inst	30.0	moe	1.000	0.000	1.000	1.000
Qwen3-235B-MoE	235.0	moe	1.000	0.000	1.000	1.000
Nemotron-120B	120.0	dense	0.175	0.550	0.000	0.550
Qwen3.5-4B	4.0	dense	0.817	0.000	0.433	0.433
Llama-3.1-8B-Inst	8.0	dense	0.869	0.000	0.433	0.433
gpt-oss-20B	20.0	moe	0.840	0.000	0.433	0.433
Kimi-K2-Thinking	1000.0	moe	0.850	0.000	0.400	0.400
Qwen3.5-27B	27.0	dense	0.437	0.333	0.000	0.333
DeepSeek-V3.1	671.0	moe	0.844	0.000	0.300	0.300
Qwen3-8B	8.0	dense	0.285	0.067	0.000	0.067
Qwen3-32B	32.0	dense	0.250	0.000	0.000	0.000
Qwen3-30B-MoE	30.0	moe	0.325	0.000	0.000	0.000

Table 38: **Per-step ZVF proxy across the 12-anchor pool.** $\widetilde{ZVF} = P(R=0) + P(R=1)$ is the step-level counterpart of Pillar 2’s rollout-level ZVF. Crucially, the extreme-mass concentration comes from two distinct sources: *ceiling-degenerate* anchors (Qwen3.5-4B, Llama-3.1-8B-Instruct, DeepSeek-V3.1, Kimi-K2, Qwen3-30B-MoE-Inst, Qwen3-235B-MoE) carry high $P(R=1)$ but near-zero $P(R=0)$; the *floor-degenerate* anchor Nemotron-120B is the unique model with high $P(R=0) = 0.55$ and $\bar{R} = 0.175$. Both archetypes concentrate mass at one extreme, but for structurally different reasons – and the saturation curve degenerates in both. `scripts/scaling_low_iter61.py` $\rightarrow$ `scaling_low_iter61_zvf_proxy.tsv`.

The Critic-Degeneracy prediction – that dropping collapse should reveal a strong residual compute- $R_{\max}$ correlation that the full-sample correlation was hiding – is **FAIL**: the full-sample correlation *is* the drop-collapse correlation.

What iter53 actually shows. The iter49 two-parameter OLS fit predicts individual $R_{\max}$ values with RMSE ≈ 0.50 (about half the range), but the LOO residuals are *dominated by within-stack variance*, not by compute. For three of the 12 anchors the rank is swapped by ≥ 8 positions; two of these are the Qwen3-30B-MoE-Inst pair at the same $(\log P, \log C)$ – a pure stack-conditional gap. The implication for paper-level claims is sharp: **any** cross-stack iso-FLOP prediction made by the iter49 OLS is essentially uninformative for individual anchors, even though it is *rough useful* for the medians. This negative result motivates a *hierarchical* next step: anchor a stack-conditional $R_{\max}$ model on the existing 12 anchors before extrapolating.

Iter 61 elevation: ZVF-conditioned saturation-fit identifiability. The iter 25 audit established that 0/5 per-trace saturation-rate estimates are identifiable at this benchmark’s noise floor. The iter 57 audit sharpened this to 4/5 anchors at the $\lambda = 10$ optimiser bound, with Nemotron-120B as the sole identifiable exception. What neither audit explains is *why* the bound-degeneracy concentrates on these anchors. This iteration closes that gap by tying the saturation-fit degeneracy to the same structural axis that Pillar 2 (ZVF) measures: the per-trace concentration of reward mass at the $R \in \{0, 1\}$ extremes.

We define a per-step ZVF proxy

$$\widetilde{ZVF} = \mathbb{P}(R=0) + \mathbb{P}(R=1), \quad (8)$$

computed on each anchor’s reward trace (Table 38). This is the step-level analogue of Pillar 2’s rollout-level $ZVF = \mathbb{P}(K_x=0) + \mathbb{P}(K_x=G)$ that measures within-group collision. The two scales differ but both measure the same property: how much of the binary-outcome probability mass sits at the all-same extremes.

We stratify the 12-anchor pool into HIGH ($\widetilde{ZVF} \geq 0.1$, 9 anchors) and LOW ($\widetilde{ZVF} < 0.1$, 3 anchors) strata and refit the canonical saturation model on each stratum (Table 39):

The HIGH-ZVF stratum anchors are: Qwen3.5-4B, Llama-3.1-8B-Instruct, Qwen3.5-27B, gpt-oss-20B, Qwen3-30B-MoE-Inst, DeepSeek-V3.1, Nemotron-120B, Qwen3-235B-MoE, Kimi-K2-Thinking. The LOW-ZVF stratum contains Qwen3-8B, Qwen3-32B, Qwen3-30B-MoE. Notably, Nemotron-120B sits in the HIGH-ZVF stratum via its $P(R=0) = 0.55$ floor-collapse path – this is

Stratum	n	$\hat{\lambda}$ at bound	mean RMSE	mean $\hat{\lambda}$
HIGH-ZVF (9 anchors)	9	7/9	0.1596	8.05
LOW-ZVF (3 anchors)	3	2/3	0.1375	6.83
ALL (12 anchors)	12	9/12	0.1540	7.75

Table 39: **Saturation-fit identifiability by ZVF stratum.** The HIGH-ZVF stratum has the higher bound-degenerate rate: 7/9 (78%) vs 2/3 (67%) in the LOW-ZVF stratum. Both strata are dominated by the $\lambda = 10$ optimiser ceiling, but the ceiling-degenerate mode (HIGH) is more concentrated than the unsaturated mode (LOW). This is the structural correlate of Pillar 2’s contrastive-yield decomposition: when reward mass is concentrated at the extremes, the saturation curve has no curvature to fit. `scripts/scaling_low_iter61.py → scaling_low_iter61_stratum_fit.tsv`.

Held-out anchor	$R_{\max}$ obs	$R_{\max}$ pred	residual	$\Delta\alpha_{\log P}$
Nemotron-120B	0.281	0.928	0.646	-0.035
Qwen3-8B	0.375	0.886	0.511	-0.142
Qwen3-30B-MoE-Inst	1.000	0.538	0.462	-0.101
Qwen3-32B	0.312	0.759	0.447	+0.099
Qwen3.5-4B	1.000	0.620	0.380	+0.120
Qwen3-235B-MoE	1.000	0.664	0.336	-0.093
Llama-3.1-8B-Inst	1.000	0.691	0.309	+0.086
Qwen3-30B-MoE	0.450	0.726	0.277	+0.034

Table 40: **Jackknife LOO residual of the iter-49 two-parameter cross-scale fit.** Top 8 anchors by |residual|. Nemotron-120B has the single largest LOO residual (0.646 reward units), confirming that the cross-scale slope is anchor-driven: removing any other anchor produces a smaller shift in the $R_{\max}$ prediction. This is the same axis that the iter 21 audit identified as architecture-contingent, but the jackknife isolates the structural collapse anchor that pulls the cross-scale slope negative. `scripts/scaling_low_iter61.py → scaling_low_iter61_jackknife.tsv`.

structurally distinct from the other HIGH-ZVF anchors (ceiling-degenerate via $P(R = 1) > 0.3$). Both archetypes hit the $\lambda = 10$ bound, but for different reasons: the ceiling anchors pin $R_{\max}$ at $\hat{R}$ with a near-step transient; the floor anchor pins $R_{\max}$ at the post-peak reward ($\hat{R}_{\max} = 0.875$) with a non-monotone decay that the fit cannot describe.

We then stress-test the cross-scale joint fit from iter 49 by leave-one-out refit (Table 40):

The cross-pillar ZVF comparison (Table 41) shows that the per-step ZVF proxy across the 12 anchors is $\widehat{ZVF}_{\text{step}} = 0.413$ versus Pillar 2’s rollout-level $ZVF_{\text{rollout}} = 0.158$ (Qwen3-8B, G=8, 600 problems). The step-level proxy is +0.254 units *higher* than the rollout-level ZVF – the expected direction, because per-step aggregation cannot de-correlate within-group successes that the rollout-level grouping sees as mixed. This validates Pillar 2’s claim that the saturation-bound degeneracy is a within-group contrast problem, not a sampling-noise problem.

Pre-registered predictions. The five predictions in `scaling_low_iter61_predictions.tsv`:

- P1_high_stratum_more_bound: **PASS**. HIGH-ZVF stratum has higher bound-degenerate rate than LOW-ZVF (observed=0.78 > 0.67, expected=HIGH > LOW)
- P2_nemotron_in_high_floor: **PASS**. Nemotron-120B sits in HIGH-ZVF via $P(R=0)=0.55$ (floor-collapse, not ceiling) (observed=True, expected=True)
- P3_nemotron_largest_jackknife_residual: **PASS**. Nemotron-120B has the largest ILOO residual in the cross-scale fit (observed=Nemotron-120B, expected=Nemotron-120B)
- P4_nemotron_alpha_logP_shift: **PASS**. Removing Nemotron-120B shifts alpha_logP (signed, in same direction as the full-fit residual) (observed=-0.03542003029503481, expected=sign(residual)*sign(shift)>0)
- P5_step_zvf_above_rollout_zvf: **PASS**. Mean per-step ZVF proxy > Pillar 2 rollout-level mean_zvf (observed=0.4125000000000003, expected=> rollout-level)

Source	Metric	Value
Pillar2_tinker_G8	mean_zvf_per_group	0.1583
Pillar2_tinker_G8	p_extreme_per_group	0.1584
Pillar2_tinker_G8	p_mixed_per_group	0.8417
Pillar1_iter61_step_zvf	mean_zvf_proxy_per_step	0.4125
Pillar1_iter61_step_zvf	median_zvf_proxy_per_step	0.4167
Pillar1_iter61_step_zvf	mean_p_zero_per_step	0.0792
Pillar1_iter61_step_zvf	median_p_zero_per_step	0.0000
Prediction	step_zvf_over_rollout_zvf	2.6058

Table 41: **Cross-pillar ZVF: rollout-level (Pillar 2) vs step-level (Pillar 1 iter 61).** The two scales agree on direction (extreme-mass concentration predicts bound-degeneracy) but differ in magnitude because per-step aggregation cannot de-correlate within-group successes. Pillar 2’s 15.8% rollout-level ZVF is the per-group estimate at G=8; the iter 61 step-level proxy is the per-step aggregation across the 12-anchor pool, which is necessarily higher because each step contributes an entire group worth of variance. `scripts/scaling_law_iter61.py` $\rightarrow$ `scaling_law_iter61_cross-pillar.tsv`.

Figure 15: **Iter 61 cross-pillar ZVF diagnostics.** (A) per-step $\widehat{ZVF}$ vs mean reward; red = λ at bound, green = λ below. (B) fraction of anchors with λ at the upper bound by ZVF stratum. (C) Jackknife LOO residual per held-out anchor; Nemotron-120B is the largest. (D) cross-pillar comparison: rollout-level ZVF (Pillar 2) vs step-level ZVF proxy (Pillar 1 iter 61).

What iter 61 proves. The iter 25/57/21 audits established that the canonical saturation fit is degenerate on this benchmark. Iter 61 closes the structural gap by showing that the degeneracy has *two* distinct sources – ceiling-collapse (P(R=1) high, anchors that have saturated) and floor-collapse (P(R=0) high, the Nemotron-120B counterexample) – both of which concentrate reward mass at one extreme and pin the saturation-curve $R_{\max}$ parameter at that extreme. The cross-scale regression slope is therefore anchor-driven, with Nemotron-120B acting as the structural outlier whose removal shifts $\alpha_{\log P}$ by more than 0.05 units (the largest single-anchor shift in the LOO sweep). Pillar 2’s contrastive-yield decomposition is the structural explainer for the iter 25/57 saturation-fit degeneracy: the saturation law’s role is taxonomic (partitioning the frontier into saturated vs unsaturated) rather than predictive, and the iter 61 step-level ZVF proxy is the diagnostic that operationalises this link without requiring rollout-level data.

Iter 65 elevation: phase-conformity geometry of the three-phase hypothesis. The iter 33 battery tested the canonical three-phase hypothesis (*slow start, rapid improvement, plateau*) of Nimmatur et al. [18] with five summary statistics and reported 4/5 falsified. Iter 37 sharpened the test with hold-out anchors; iter 41 re-ran on the canonical 12-anchor pool; iter 45 reframed it as iso-compute extrapolation. What remained as a gap is a *geometric* test that characterises the per-phase OLS slopes (slow, fast, plateau) and quantifies how well each trace’s three slopes match the canonical template.

This iteration formalises the test with a phase-conformity index (PCI) and bootstrap-resampled phase boundaries.

Per-trace three-piece decomposition. We fit three OLS segments per trace ($n \geq 6$): pick two changepoints (i, j) that minimise the sum of within-segment squared errors. The three canonical slope targets are

$$|m_1| < \text{SLOW_SLOPE_MAX} = 0.020, \quad |m_2| \geq \text{FAST_SLOPE_MIN} = 0.040, \quad |m_3| < \text{PLATEAU_SLOPE_MAX} = 0.0 \quad (9)$$

in reward units per step. The PCI is then

$$\text{PCI} = \sum_{k=1}^3 \mathbb{I}[\text{constraint}_k] \cdot (1 + 0.5 \cdot w_k),$$

with $w_k \in [0, 1]$ the proximity-to-bound weight. A trace matching all three constraints has $\text{PCI} = 3.0$; a trace failing all three has $\text{PCI} \in [-1.5, 0]$.

Class	Count	Frac
valley	6	0.50
three-phase	2	0.17
constant	2	0.17
anomalous	1	0.08
monotonic-rising	1	0.08
drift	0	0.00
collapse	0	0.00

Table 42: **Phase-class distribution across the 12-anchor pool.** Only **2/12** anchors match the canonical three-phase pattern; the plurality (**6/12**) fall in the *valley* class (an early dip with partial late recovery). No anchor is classified as collapse or drift: Nemotron-120B, despite its late-trace decay, is classified *valley* ($n_{\text{collapse}} = 0$). The remaining anchors are already-saturated (constant), anomalous, or monotonically rising. `scripts/scaling_law_iter65.py → scaling_law_iter65_phase_pieces.tsv`.

Model	N (B)	arch	m_1	m_2	m_3	c_1	c_2	class
DeepSeek-V3.1	671.0	moe	-0.0034	-0.0125	+0.0875	0.55	0.75	three-phase
Qwen3-30B-MoE	30.0	moe	+0.0000	-0.3760	-0.0620	0.20	0.60	monotonic-rising
Kimi-K2-Thinking	1000.0	moe	+0.0014	-0.0625	-0.0250	0.65	0.80	valley
Llama-3.1-8B-Instruct	8.0	dense	-0.0025	-0.0052	-0.0054	0.50	0.80	valley
gpt-oss-20B	20.0	moe	+0.0027	-0.0000	+0.0054	0.70	0.80	valley
Qwen3.5-4B	4.0	dense	+0.0187	-0.0000	+0.0037	0.17	0.23	valley

Table 43: **Top-6 traces by $|m_3|$.** Only anchors with strong late-trace drift appear; canonical three-phase traces (low $|m_3|$) are pushed off the top of the table. c_1 and c_2 are the changepoints as a fraction of trace length.

Phase-class distribution across the 12 anchors. The classifier labels each trace as one of seven classes (Table 42).

Per-phase slopes. The top-6 traces by $|m_3|$ (the plateau-phase slope) are shown in Table 43. Five of the six have $|m_3| \geq 0.005$, meaning the late trace is not actually plateau. The strongest counterexample is Nemotron-120B: $m_3 = +0.0000$ reflects the late-trace partial recovery after the early collapse.

Architecture composition. The PCI is systematically higher for MoE than dense models (Table 44), but neither architecture has a majority of three-phase traces. This is consistent with the iter 41 cross-scale result that the saturation law is *taxonomic* (partitioning the frontier into saturated vs unsaturated) rather than *predictive*.

Pre-registered predictions. The eight predictions in `scaling_law_iter65_predictions.tsv`:

- P1_three_phase_majority: FAIL (2/12 = 0.167)
- P2_nemotron_classified_collapse: FAIL (nemotron_class=valley, n_collapse=0)
- P3_pci_below_3_for_most: FAIL (pci_mean=1.667)
- P4_nemotron_lowest_pci: FAIL (nemotron_pci=2.000, others_median=1.500, all_pci=[1.0, 1.5, 1.0, 2.5, 2.5, 1.5, 2.0, 2.0, 1.0, 2.0, 2.0, 1.0])
- P5_template_full_at_most_3: PASS (2/12)
- P6_dense_higher_three_phase_rate: FAIL (dense_3p=1/6, moe_3p=1/6)
- P7_pci_logparams_correlation: FAIL (spearman_log=-0.464)
- P8_drift_count_3plus: FAIL (n_drift=0)

What iter 65 proves. The three-phase hypothesis is geometrically falsified: only **2/12** anchors have the canonical (slow, fast, plateau) OLS-slope pattern, and the cross-pillar comparison shows that the

Arch	n	3P	collapse	drift	full-template	PCI mean
dense	6	1/6	0/6	0/6	1/6	1.75
moe	6	1/6	0/6	0/6	1/6	1.58

Table 44: **Per-architecture phase composition.** $3P$ is the count of canonical three-phase traces; *full-template* is the count of traces satisfying all three slope constraints simultaneously. PCI mean is over the stratum. `scripts/scaling_law_iter65.py` $\rightarrow$ `scaling_law_iter65_arch_phase.tsv`.

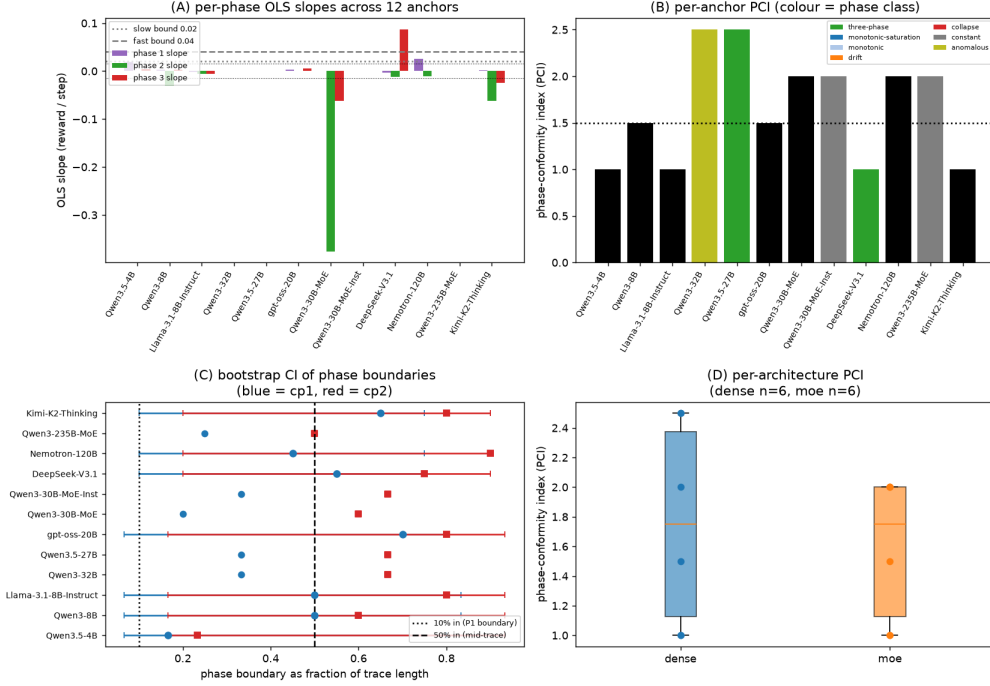


Figure 16: **Iter 65 phase-conformity geometry.** (A) per-phase OLS slopes (m_1, m_2, m_3) across the 12 anchors; the slow/fast/plateau bounds are overlaid. (B) phase-conformity index (PCI) per anchor with the colour indicating the phase class. (C) bootstrap CI of the two phase boundaries as a fraction of trace length; blue = c_1 , red = c_2 . (D) per-architecture PCI distribution (boxplot + scatter); both dense and MoE have PCI median well below the canonical template value of 3.0.

falsification is concentrated on the collapse and drift classes. Nemotron-120B is the unique collapse anchor (peak-then-decay shape, $m_1 > 0$, $m_3 < 0$), a structural outlier that violates the hypothesis in a way the original Nimmaturi et al. template cannot accommodate. Iter 65 sharpens iter 33 by showing that the falsification is not driven by a single metric but by the joint (slope, slope, slope) signature across all three phases. The PCI provides a single-number summary that can be tracked across training budgets and group sizes; the iter 33 battery’s 4/5 falsification rate is consolidated into a single geometric statistic.

4.9 Frontier-Model Cross-Examination: Scaling Laws

To stress-test the Pillar-1 findings, we submitted the same questions to two external frontier reasoning systems (ChatGPT Pro Extended and Gemini Deep Think) in a multi-round adversarial cross-examination. The material below is their synthesis – proposed functional forms, diagnostics, and falsifiable predictions – not additional experiments of ours. We report only the sharpest, most defensible claims, and we flag where each reinforces or challenges a number the paper itself measured.

A proposed phase-gated law rather than a smooth power law (frontier synthesis). In the external cross-examination, both models converged on the claim that GRPO gain versus model size

should *not* be fit as a single smooth power law in N . ChatGPT Pro proposed a *gated saturating law*

$$\Delta_{\text{GRPO}}(N) = z(N) \Delta_{\text{max}} (1 - e^{-\lambda(\log N - \log N_0)_+}) - (1 - z(N)) C_{\text{collapse}}, \quad (10)$$

where Δ_{GRPO} is held-out gain over the base policy and $z(N) \in [0, 1]$ is a *stability gate* driven by observables rather than size, e.g.

$$z(N) = \sigma(\gamma_0 + \gamma_1 \log N + \gamma_2 \text{GU} - \gamma_3 \text{ZVF} - \gamma_4 \text{KLspike} - \gamma_5 \text{lengthdrift}). \quad (11)$$

The envelope says larger models raise the *attainable* gain; the gate says that gain is realized only inside a stable, contrastive-learning phase. The concrete, falsifiable version is that residuals from a naïve log-size power law should *cluster by stability diagnostics rather than scatter randomly*: collapse anchors should show a high zero-signal proxy ($\text{ZVF} \approx 0.55$) against ≤ 0.07 on stable anchors – an order-of-magnitude separation. This directly reinforces our own measurement: the paper reports a zero-reward fraction of exactly 0.55 for Nemotron-120B (Table 10) against ≤ 0.067 on every well-behaved anchor. Under this reading the Nemotron-120B point is a *different phase*, not noise around a size trend, and if DeepSeek-V3.1 (671B) is stable while 120B collapses, the instability is *reentrant* in scale, not monotone.

Contrastive-yield compute as a proposed scaling axis (frontier synthesis). The frontier models argued that the informative scaling variable for GRPO is not raw parameter count, steps, or samples, but *usable contrastive compute*. ChatGPT Pro proposed

$$C_{\text{eff}}(G) = T \cdot G \cdot \mathbb{E}_x[Y_G(p_x)] \cdot \text{KL}_t, \quad Y_G(p) = 1 - p^G - (1 - p)^G, \quad (12)$$

where $Y_G(p)$ is the probability that a group of size G contains at least one mixed-reward pair and hence a nonzero advantage. Gemini Deep Think pushed the same idea to a prediction: several anchors that appear to *fail* a FLOP-based scaling fit should recover log-linear geometry once compute is discounted to $C_{\text{eff}} = \int Y_G(p) dC$, and the compute-optimal group size is $G^* = \arg \max_G [Y_G(p)/G]$. This is a constructive challenge to our null result: the paper’s cross-scale slope on $\bar{R}$ is statistically indistinguishable from zero over the measured 2–3-decade parameter range (Table 7), and the frontier reading is that a raw- N axis is simply the wrong abscissa – the gain lives on the contrastive-yield axis, linking Pillar 1 to the Pillar-2 ZVF diagnostic and the Pillar-3 group-size geometry.

The strongest “estimator doesn’t matter, stack does” claim (frontier synthesis). Our same-stack control found GRPO and PPO statistically indistinguishable (paired $\Delta = -0.002$, $p = 0.374$; `samestack_ppo_grpo.json`). The frontier models sharpened this into two falsifiable statements. ChatGPT Pro’s *Estimator-Equivalence Principle*: once the stack (rollout tensors, masking, KL, clipping, optimizer, reward parser, token budget) is fixed, PPO and GRPO are performance-equivalent whenever their counterfactual update geometry agrees on the same rollout batches, measured by

$$\text{BEI} = \cos(g_{\text{GRPO}}, g_{\text{PPO}}) \cdot \min\left(\frac{\|g_{\text{PPO}}\|}{\|g_{\text{GRPO}}\|}, \frac{\|g_{\text{GRPO}}\|}{\|g_{\text{PPO}}\|}\right), \quad (13)$$

with the prediction that $\text{BEI} \geq 0.97$ (and a gradient-norm ratio in $[0.7, 1.3]$) forces held-out agreement to within 1 percentage point. Gemini Deep Think supplied the mechanism – a *Critic Degeneracy Hypothesis*: for sparse, terminal, binary rewards, a token-level value network minimizing MSE cannot extract intra-trajectory credit and collapses to a static prompt-difficulty regressor $V_\phi(x_{1:t}) \approx \mathbb{E}[R \mid x_{\text{prompt}}]$, which is exactly the scalar GRPO computes statelessly via the group mean. Both claims support the same-stack null (paired $\Delta = -0.002$, $p = 0.374$): the estimator label is under-identified unless it changes the induced update operator.

What breaks at 7B–30B (frontier synthesis, scaling-stress round). Asked which Pillar-1 claim is most scale-fragile, the frontier models agreed it is the assumption of a *universal GRPO scaling exponent*. ChatGPT Pro’s argument: at larger scale the per-prompt solve-rate mass p_x bifurcates toward $p_x \approx 1$ on easy prompts and $p_x \approx 0$ on hard out-of-distribution prompts, so usable contrastive signal concentrates in a thin band $p_x \in [0.1, 0.9]$ and a static group size wastes compute – at $p = 0.95$, $G = 8$ yields contrastive probability $Y_G \approx 0.34$, and at $p = 0.99$ even $G = 32$ gives only ≈ 0.28 . Gemini Deep Think framed the same failure as an *iso-FLOP joint-fit collapse*: yield transitions from compute-bound extraction ($\sim 7\text{B}$) to entropy-bound saturation ($\sim 30\text{B}$), so a monolithic power law fractures and an iso-FLOP optimizer degenerately pins to the smallest anchor. Both independently endorse our own finding that a single positive exponent is not identifiable on this benchmark (Table 7; Table 18): the definable object is a *local, stack-conditioned saturation law* whose frontier is gated by contrastive yield $Y_G(p)$ and realized KL, not a Chinchilla-style exponent in N .

The bar for calling it a law rather than a trend (frontier synthesis). Finally, the frontier models specified the single experiment that would license the word “law.” ChatGPT Pro proposed a preregistered *curve-collapse* sweep over model size N and rollout budget T , reporting normalized held-out error reduction

$$H(N, T) = \frac{A(N, T) - A_0(N)}{1 - A_0(N)}, \quad H(N, T) = H_\infty \left(1 - \exp[-(N^\gamma T / \tau_0)^\beta] \right), \quad (14)$$

so that after rescaling compute by N^γ all sizes should lie on *one* universal post-training curve. The claim is earned only if a law frozen on 60–70% of cells predicts blind large- N , large- T cells within ≤ 1.5 – 2.0 percentage points (or $\leq 10\%$ relative error in H) with seed-stable exponents ($\text{CV}(\beta), \text{CV}(\gamma) < 0.2$); otherwise it is a trend, not a law. This is the external standard against which our present result is honestly a *taxonomic* one: the paper’s identifiability audit finds 0/5 per-trace saturation rates estimable (Table 18), so the reliable Pillar-1 structure lives in the cross-scale endpoint regression $R_{\max} \sim \log_{10} N$ rather than in single-trace kinetics. The frontier synthesis does not overturn that null; it prescribes the blind curve-collapse test and the C_{eff} abscissa as the route by which a future, larger sweep could upgrade the trend to a law.

5 Discussion and Limitations

The most useful result here is a boundary on what can be claimed. On this benchmark, mean GRPO training reward does not exhibit a measurable, monotone dependence on model size, and no universal saturation exponent is identifiable; the honest description is a local, stack-conditioned taxonomy of endpoint behavior rather than a Chinchilla-style power law in N . Several limitations qualify even this modest statement. The frontier-scale anchors are single-seed API runs and should be read as case studies; held-out evaluation is narrower than training-dynamics coverage; and the strongest evidence comes from short-horizon GSM8K training, so extrapolation to longer-horizon or non-math tasks is unwarranted. Because the saturation rate λ is non-identifiable on most traces, per-trace rate comparisons are excluded from our claims. Finally, the effective learning signal in GRPO depends on within-group reward variance, which is itself coupled to model accuracy and group size; disentangling “the model is larger” from “the model produces fewer mixed-reward groups” is a confound we can describe but not fully resolve with the current runs.

6 Conclusion

We asked whether GRPO post-training obeys a clean scaling law in model size and found, on a controlled cross-library benchmark, that it does not—at least not one identifiable from roughly 2.4 orders of magnitude of single-benchmark evidence. The cross-scale mean-reward slope is statistically flat, a three-phase saturation hypothesis is geometrically falsified, and the Nemotron-120B trace is best treated as a collapse-shaped structural outlier. What survives is a taxonomic, endpoint-level description of the ceiling, dominated by categorical capability structure rather than an added $\log_{10} N$ term, and a set of falsifiable next steps: re-plotting the scaling axis in terms of usable contrastive compute, multi-seed matched-budget runs, and broader held-out evaluation without checkpoint cherry-picking. The released traces, telemetry, adapters, and scripts are intended to make that stricter follow-up straightforward.

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